

LIU, Yang | Curriculum Vitae

NANYANG TECHNOLOGICAL UNIVERSITY

Full Professor (with Tenure)
President's Chair Professor in Computer Science and Engineering
Executive Director for CyberSG Research&Development Programme Office (CRPO)
Executive Director for Cyber Security Research Centre @ NTU (Cysren)
Director for Cyber Security Lab, School of Computer Science and Engineering
Programme Director for HP-NTU Digital Manufacturing Corp Lab
Trust Lead of the Continental-NTU Corp Lab
Cluster Director in Cybersecurity, Energy Research Institute @NTU

Education

National University of Singapore <i>Ph.D., CAP 4.56/5</i> Thesis: Model Checking Concurrent and Real-time Systems: the PAT Approach	Singapore <i>Aug 2005–May 2010</i>
National University of Singapore <i>Bachelor of Computing Science, Mathematics Minor, CAP 4.83/5</i> First Class Honour	Singapore <i>Aug 2001–Jun 2005</i>

Working Experience

School of Computer Engineering, Nanyang Technological University <i>Full Professor</i>	Singapore <i>Sep 2019–Present</i>
School of Computer Engineering, Nanyang Technological University <i>Associate Professor</i>	Singapore <i>Sep 2017–Aug 2019</i>
School of Computer Engineering, Nanyang Technological University <i>Nanyang Assistant Professor</i>	Singapore <i>Oct 2012–Aug 2017</i>
Temasek Laboratories, National University of Singapore <i>Senior Research Scientist</i>	Singapore <i>Aug 2011–Sep 2012</i>
Singapore University of Technology and Design <i>Post Doc Research Fellow</i>	Singapore <i>April 2011–July 2011</i>
Massachusetts Institute of Technology <i>Visiting Scholar</i>	USA <i>Feb 2011–March 2011</i>
School of Computing, National University of Singapore <i>Post Doc Research Fellow</i>	Singapore <i>May 2010–Jan 2011</i>
Theory Group, Microsoft Research Asia <i>Microsoft Research Asia Fellow</i>	Beijing, China <i>Feb 2008–Aug 2008</i>
National University of Singapore <i>Research Assistant</i>	Singapore <i>Nov 2005–May 2006</i>

RESEARCH

(1) Awards.....

Software Research Prototype System Competition 1st Prize Winner **China**
ChinaSoft 2025 **2025**

IntelliRadar: A Comprehensive Platform to Pinpoint Malicious Package Information from Cyber Intelligence,
The 2025 National Software Application Conference, China

Distinguished Paper Award **Trondheim, Norway**
FSE (Top Tier Conference) **June 2025**

Xiaokun Luan, David Sanan, Zhe Hou, Qiyuan Xu, Chengwei Liu, Yufan Cai, Yang Liu, Meng Sun. *Why the Proof Fails in Different Versions of Theorem Provers: An Empirical Study of Compatibility Issues in Isabelle*. ACM International Conference on the Foundations of Software Engineering (FSE 2025), Trondheim, Norway, June 23-27, 2025.

Top Cited Article Award **WILEY**
Journal of Software: Evolution and Process **April 2025**

Huangzhao Zhang, Shuai Lu, Zhuo Li, Zhi Jin, Lei Ma, Yang Liu, Ge Li. *CodeBERT-Attack: Adversarial attack against source code deep learning models via pre-trained model*. *Journal of Software: Evolution and Process*, vol. 36, no. 3, 2023.

Public Sector Cybersecurity Champion 2025 **GovInsider**
Journal of Software: Evolution and Process **April 2025**

Best Industry Challenge Paper Award **Ottawa, Canada**
ICSE (Top Tier Conference) **April 2025**

Hanxiang Xu, Wei Ma, Ting Zhou, Yanjie Zhao, Kai Chen, Qiang Hu, Yang Liu, Haoyu Wang. *CKGFuzzer: LLM-Based Fuzz Driver Generation Enhanced By Code Knowledge Graph*. IEEE/ACM International Conference on Software Engineering (ICSE 2025), Ottawa, Canada, April 27-May 3, 2025.

Distinguished Paper Award **San Diego, CA, USA**
NDSS (Top Tier Conference) **February 2025**

Ye Liu, Yue Xue, Daoyuan Wu, Yuqiang Sun, Yi Li, Miaolei Shi, Yang Liu. *PropertyGPT: LLM-driven Formal Verification of Smart Contracts through Retrieval-Augmented Property Generation*. Network and Distributed System Security Symposium (NDSS 2025), San Diego, California, USA, February 24-28, 2025.

Distinguished Artifact Award **Salt Lake City, USA**
CCS (Top Tier Conference) **October 2024**

Dongqi Han, Zhiliang Wang, Ruitao Feng, Minghui Jin, Wenqi Chen, Kai Wang, Su Wang, Jiahai Yang, Xingang Shi, Xia Yin, Yang Liu. *Rules Refine the Riddle: Global Explanation for Deep Learning-Based Anomaly Detection in Security Applications*. ACM Conference on Computer and Communications Security (CCS 2024), Salt Lake City, USA, October 14-18, 2024.

Distinguished Artifact Award **Philadelphia, PA, USA**
USENIX (Top Tier Conference) **August 2024**

Gelei Deng, Yi Liu, Víctor Mayoral-Vilches, Peng Liu, Yuekang Li, Yuan Xu, Tianwei Zhang, Yang Liu, Martin Pinzger, Stefan Rass. *PentestGPT: Evaluating and Harnessing Large Language Models for Automated Penetration Testing*. ACM Conference on USENIX Security (USENIX 2024), August 14–16, 2024, Philadelphia, PA, USA.

Distinguished Computing Alumni Award 2023
National University of Singapore **June 2024**
School of Computing.

ACM SIGSOFT Distinguished Paper Award**Porto de Galinhas, Brazil***FSE (Top Tier Conference)**June 2024*

Kaixuan Li, Yue Xue, Sen Chen, Han Liu, Kairan Sun, Ming Hu, Haijun Wang, Yang Liu, Yixiang Chen. *Static Application Security Testing (SAST) Tools for Smart Contracts: How Far Are We?*. 2024 ACM International Conference on the Foundations of Software Engineering (FSE 2024), July 17-19, 2024, Porto de Galinhas, Brazil.

Best Paper Award**Singapore***Continental-NTU Corporate Lab**Apr 2024*

Cen Zhang and Yuekang Li, Hao Zhou, Xiaohan Zhang, Yaowen Zheng, Xian Zhan, Xiaofei Xie, Xiapu Luo, Xinghua Li, Yang Liu and Sheikh Mahbub Habib. *Automata-Guided Control-Flow-Sensitive Fuzz Driver Generation*. The 32th USENIX Security Symposium, 2023.

Best Patent Award**Singapore***Continental-NTU Corporate Lab**Apr 2024*

JingQuan Ge, Sheikh Mahbub Habib, Kakkolangara Praveen, Yang Liu. *Computer-readable Data Carrier, Apparatus, Computer Program and Method for Fuzz Testing Software*

ACM SIGSOFT Distinguished Paper Award**Lisbon, Portugal***ICSE (Top Tier Conference)**Feb 2024*

Yiming Liu, Cen Zhang, Feng Li, Yeting Li, Jianhua Zhou, Jian Wang, Lanlan Zhan, Yang Liu, Wei Huo. *Semantic-Enhanced Static Vulnerability Detection in Baseband Firmware*. 2024 IEEE/ACM 46th International Conference on Software Engineering (ICSE 2024), April 14–20, 2024, Lisbon, Portugal.

Distinguished Paper Award**California***NDSS 2024 Workshop on AISCC**Feb 2024*

Gelei Deng, Yi Liu, Kailong Wang, Yuekang Li, Tianwei Zhang, Yang Liu. *Pandora: Jailbreak GPTs by Retrieval Augmented Generation Poisoning*. The Network and Distributed System Security Symposium (NDSS 2024) Workshop on Artificial Intelligence System with Confidential Computing (AISCC 2024).

TSE Best Paper Award**IEEE***TSE (Top Tier Journal)**Dec 2023*

Jie M. Zhang, Mark Harman, Lei Ma, Yang Liu. *Machine Learning Testing: Survey, Landscapes and Horizons*. IEEE Transactions on Software Engineering, 2022

ACM SIGSOFT Distinguished Paper Award**Luxembourg***ASE 2023 (Top Tier Conference)**Sep 2023*

Kai Huang, Xiangxin Meng, Jian Zhang, Yang Liu, Wenjie Wang, Shuhao Li, Yuqing Zhang. *An Empirical Study on Fine-tuning Large Language Models of Code for Automated Program Repair*. The 38th IEEE/ACM International Conference on Automated Software Engineering

ACM SIGSOFT Distinguished Paper Award**Australia***ICSE 2023 (Top Tier Conference)**Feb 2023*

Lyuye Zhang, Chengwei Liu, Zhengzi Xu, Sen Chen, Lingling Fan, Lida Zhao, Jiahui Wu and Yang Liu. *Compatible Remediation on Vulnerabilities from Third-Party Libraries for Java Projects*. The 45rd International Conference on Software Engineering

Best Paper Award**Israel***ECCV-AROW 2022**Oct 2022*

Liangru Sun, Felix Juefei-Xu, Yihao Huang, Qing Guo, Jiayi Zhu, Jinciao Feng, Yang Liu, Geguang Pu. *ALA: Adversarial Lightness Attack via Naturalness-aware Regularizations* ECCV 2022 Workshop on Adversarial Robustness in the Real World

ACM SIGSOFT Distinguished Paper Award**USA***ASE 2022 (Top Tier Conference)**Oct 2022*

Lyuye Zhang, Chengwei Liu, Zhengzi Xu, Sen Chen, Lingling Fan, Bihuan Chen, Yang Liu. *Has My Release Disobeyed Semantic Versioning? Static Detection Based On Semantic Differencing* The 37th IEEE/ACM International Conference on Automated Software Engineering

ACM SIGSOFT Distinguished Paper Award **South Korea**
ISSTA 2022 (Top Tier Conference) *July 2022*
 Zhiming Li, Xiaofei Xie, Haoliang Li, Zhengzi Xu, Yi Li and Yang Liu. *Cross-Lingual Transfer Learning for Statistical Type Inference*. The 31st International Symposium on Software Testing and Analysis

CoE Innovator (Entrepreneurship) Award **Singapore**
Nanyang Technological University *June 2022*

Most Influential Research Paper of Year 2021
Ants Finance *April 2022*
 Cen Zhang, Xingwei Lin, Yuekang Li, Yinxing Xue, Yang Liu. *APICraft: Fuzz Driver Generation for Closed-source SDK Libraries*.

3rd Place of AI Singapore's Trusted Media Challenge with S\$100K Price **Singapore**
AI Singapore *Feb 2022*

Most Innovative Paper Award **USA**
ISPA 2021 *October 2021*
 Xingshuo Han, Kangjie Chen, Yuan Zhou, Meikang Qiu, Chun Fan, Yang Liu and Tianwei Zhang. *A Unified Anomaly Detection Methodology for Lane-Following of Autonomous Driving Systems*. IEEE International Symposium on Parallel and Distributed Processing with Applications

ACM SIGSOFT Distinguished Paper Award **Spain**
ICSE 2021 *May 2021*
 Xian Zhan, Lingling Fan, Sen Chen, Feng Wu, Tianming Liu, Xiapu Luo and Yang Liu. *ATVHunter: Reliable Version Detection of Third-Party Libraries for Vulnerability Identification in Android Apps*. The 43rd International Conference on Software Engineering

Vulnerability Detection Tool Competition 3rd Prize **China**
ChinaSoft 2020 *Nov 2020*
 The 2020 CCF ChinaSoft Conference, China

IEEE TCSE Distinguished Paper Award **Australia**
ICSME 2020 *Sep 2020*
 Ying Wang, Bihuan Chen, Kaifeng Huang, Bowen Shi, Congying Xu, Xin Peng, Yijian Wu and Yang Liu. *An Empirical Study of Usages, Updates and Risks of Third-Party Libraries in Java Projects*. The 36th The International Conference on Software Maintenance and Evolution

ACM's Distinguished Speaker
Association for Computing Machinery *18 March 2020*

NRF Investigatorship, Class 2020 **Singapore**
National Research Foundation (NRF) *Jan 2020*
 The NRF Investigatorship provides opportunities for established, innovative and active scientists and researchers, in their mid-career, to pursue ground-breaking, high-risk research. It is designed to support a small number of excellent Principal Investigators who have a track record of research achievements that identify them as leaders in their respective field(s) of research.

ACM Distinguished Paper Award **San Diego, USA**
ASE 2019 (Top Tier Conference) *Nov 2019*
 Yan Zheng, Xiaofei Xie, Ting Su, Lei Ma, Jianye Hao, Zhaopeng Meng, Yang Liu, Ruimin Shen, Yinfeng Chen and Changjie Fan. *Wuji: Automatic Online Combat Game Testing Using Evolutionary Deep Reinforcement Learning*. The 34th IEEE/ACM International Conference on Automated Software Engineering

Research Tool Award 1st Price with 2000SGD Cash Award **China**
NASAC 2019 *2019*
BiFF: An Effective Binary Fuzzing Framework with Cross-Architecture Support, The 18th National Software Application Conference, China

Research Tool Award 2nd Price with 1000SGD Cash Award **China**
NASAC 2019 **2019**
DeepStudio: a Framework of Securing Deep Learning Systems, The 18th National Software Application Conference, China

Nanyang Research (Young Investigator) Award **Singapore**
Nanyang Technological University **28 Aug 2019**

2019 University Leadership Forum Chair **Singapore**
Nanyang Technological University **1 April 2019 - 31 Mar 2024**

The 20 Year Most Influential System Award **Gold Coast, Australia**
ICFEM 2018 **15 November, 2018**
 Jinsong Dong, Yang Liu and Jun Sun. *PAT Verification System*. The 20th International Conference On Formal Engineering Methods

ACM Distinguished Paper Award **Montpellier, France**
ASE 2018 (Top Tier Conference) **3 to 7 September, 2018**
 Lei Ma, Felix Juefei-Xu, Fuyuan Zhang, Jiyuan Sun, Minhui Xue, Bo Li, Chunyang Chen, Ting Su, Li Li, Yang Liu, Jianjun Zhao and Yadong Wang. *DeepGauge: Multi-Granularity Testing Criteria for Deep Learning Systems*. The 33rd IEEE/ACM International Conference on Automated Software Engineering

ACM Distinguished Paper Award **Montpellier, France**
ASE 2018 (Top Tier Conference) **3 to 7 September, 2018**
 Kaifeng Huang, Bihuan Chen, Xin Peng, Daihong Zhou, Ying Wang, Yang Liu, and Wenyun Zhao. *CLDIFF: Generating Concise Linked Code Differences*. The 33rd IEEE/ACM International Conference on Automated Software Engineering

Best Paper Award **Beijing, China**
Internetwork 2018 **September, 2018**
 Lingyun Situ, Linzhang Wang, Yang Liu, Bing Mao, Xuandong Li. *Vanguard: Detecting Missing Checks for Prognosing Potential Vulnerabilities*. the Tenth Asia-Pacific Symposium on Internetwork

ACM Distinguished Paper Award **Gothenburg, Sweden**
ICSE 2018 (Top Tier Conference) **May 27-3 June, 2018**
 Lingling Fan, Ting Su, Sen Chen, Guozhu Meng, Yang Liu, Lihua Xu, Geguang Pu and Zhendong Su. *Large-Scale Analysis of Framework-Specific Exceptions in Android Apps*. The 40th International Conference on Software Engineering

Tan Chin Tuan Fellowship 2018 **Singapore**
Nanyang Technological University **2018**

ACM Distinguished Paper Award **Illinois, USA**
ASE 2017 (Top Tier Conference) **October 30 - November 3, 2017**
 Shang-Wei Lin, Jun Sun, Hao Xiao, Yang Liu, David Sanan and Henri Hansen. *FiB: Squeezing Loop Invariants by Interpolation between Forward/Backward Predicate Transformers*. The 32nd IEEE/ACM International Conference on Automated Software Engineering

Best Research Tool Award with 3000SGD Cash Award **China**
NASAC 2017 **2017**
Guided, Stochastic Model-Based GUI Testing of Android Apps, The 16th National Software Application Conference, China

First Place in Vulnerability Hunting Competition with 5000SGD Cash Award **China**
NASAC 2017 **2017**
 The 16th National Software Application Conference, China

Best Paper Award Finalist**Austin, USA***DAC 2017 (Top Tier Conference)**June 18-22, 2017*

Wenjian He, Sanjeev Das, Wei Zhang and Yang Liu. *No-Jump-into-Basic-Block: Enforce Basic Block CFI on the Fly for Real-world Binaries*. IEEE/ACM Design Automation Conference (DAC)

Best Paper Award in Automation Finalist**Singapore***ICRA 2017 (Top Tier Robotic Conference)**June 2017*

Yan Yang, Hesuan Hu and Yang Liu. *A distributed approach to automated manufacturing systems with complex structures using Petri nets*. IEEE International Conference on Robotics and Automation

ACM Distinguished Paper Award**Seattle, USA***FSE 2016 (Top Tier Conference)**November 13-18, 2016*

Xiaofei Xie, Bihuan Chen, Yang Liu, Wei Le and Xiaohong Li. *Proteus: Computing Disjunctive Loop Summary via Path Dependency Analysis*. ACM SIGSOFT International Symposium on the Foundations of Software Engineering

Best Paper Award**Paris, France***IFIPTM 2016**July 18-22, 2016*

Tim Muller, Dongxia Wang, Jie Zhang and Yang Liu. *How to Use Information Theory to Mitigate Unfair Rating Attacks*. 10th IFIP WG 11.11 International Conference on Trust Management (IFIPTM 2016), July 18-22, Darmstadt, Germany, 2016

Tan Chin Tuan Fellowship 2016**Singapore***Nanyang Technological University**2016***Best SoSyM Journal Paper Award****Ottawa, Canada***Model 2015**September 30 - Oct 2, 2015*

Songzheng Song, Jiexin Zhang, Yang Liu, Mikhail Auguston, Jun Sun, Jin Song Dong, Tieming Chen. *Formalizing and Verifying Stochastic System Architectures Using Monterey Phoenix*. ACM/IEEE 18th Intl. Conf. on Model Driven Engineering Languages and Systems. (Model 2015).

Nanyang Assistant Professorship**Singapore***Nanyang Technological University**May 2013*

The Nanyang Assistant Professorship (NAP) scheme was launched in 2007 as an elite early career award for outstanding young researchers and exceptional scholars worldwide who aspire to research leadership role at NTU. NAP's awardees receive S\$1 million of research start-up grants to establish their research in NTU.

Best Paper Award**Paris, France***ICECCS 2012**July 18th - 20th, 2012*

Yan Liu, Xian Zhang, Yang Liu, Jun Sun, Jin Song Dong, Jit Biswas and Mounir Mokhtari. *Formal Analysis of Pervasive Computing Systems*. The 17th IEEE International Conference on Engineering of Complex Computer Systems (ICECCS 2012), Paris, France, July 18 - 20, 2012

Temasek Research Fellowship 2011**Singapore***Defence Research and Technology**2011*

The Temasek Research Fellowship is a scheme aimed at recruiting outstanding young researchers with a Ph.D Degree in science or technology. The scheme provides young researchers an opportunity to conduct and lead research that is relevant to defence.

Best Paper Award**Singapore***SSIRI 2010**June 2010*

Shaojie Zhang and Yang Liu. *Model Checking a Lazy Concurrent List-Based Set Algorithm*. The 4th IEEE International Conference on Secure Software Integration and Reliability Improvement (SSIRI 2010), pp. 43-52, Singapore, June 2010.

Dean's Graduate Research Excellence Award <i>National University of Singapore</i> The prestigious award is given to senior PhD students who have made significant research achievements during their PhD study. Recipients of this award must produce strong evidence of sustained research achievement during their course of study. Deserving recipients will receive this award once throughout their candidature.	Singapore <i>Nov 2010</i>
Research Achievement Award <i>National University of Singapore</i> It will be presented to PhD students who have achieved outstanding research performance over the past academic year.	Singapore <i>Nov 2009</i>
Microsoft Asia Research Fellowship <i>Microsoft Asia Research</i> The annual Microsoft Fellowship Program is designed to support talented PhD students who have the potential to become future research leaders. Over the past seven years, 159 recipients have participated in the program. Fellowships have been awarded to outstanding students from more than 30 universities and research institutes throughout the Asia-Pacific Region since 1999.	Beijing, China <i>Aug 2007</i>
Graduate Scholarship <i>National University of Singapore</i> Graduate Scholarship in National University of Singapore	Singapore <i>Aug 2005 - Nov 2009</i>
Tata Consultancy Services Asia Pacific Medal and Prize <i>National University of Singapore</i> The second best graduate throughout the course of study for the Bachelor of Computing (Honours) in the School of Computing, NUS. (2nd of 800+ students with CAP 4.83/5).	Singapore <i>July 2005</i>
Undergraduate Scholarship <i>MOE of Singapore</i> Undergraduate Scholarship for PRC Students	Singapore <i>Dec 2000 - June 2005</i>
(2) Citations	
Total number of citations: 42334 (Google Scholar as of Feb 2026)	
H-index = 100 (H-index is the maximum value of H such that there are H papers co-authored by me with H or more citations.)	
(3) Research Organisations	
ICECCS 2025 <i>General Chair</i> The International Conference on Engineering of Complex Computer Systems	Hang Zhou <i>July 2025</i>
FSE 2024 Student Research Competition <i>PC Chair</i> The 32th ACM International Conference on the Foundations of Software Engineering	Brazil <i>July 2024</i>
PRDC 2023 <i>General Co-Chair</i> The 28th IEEE Pacific Rim International Symposium on Dependable Computing	Singapore <i>Nov 2023</i>
FSE 2022 <i>Artifact Evaluation Co-Chair</i> The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)	Singapore <i>Nov 2022</i>
ASE 2022 <i>PC member</i> 37th IEEE/ACM International Conference on Automated Software Engineering	United States <i>October 2022</i>

SEAIS 2022 <i>General Co-Chair</i> The 1st International Conference on Linking Software Engineering, Artificial Intelligence and Security	Singapore March 2022
I-Day@FM2021 <i>Program Co-Chair</i> Industry Day of Formal Methods 2021	China November 2021
NSS 2021 <i>Program Co-Chair</i> The 15th International Conferences on Networks and Systems Security	China Oct 2021
Then 12th Internetwork Conference <i>General Chair</i> The 12th Internetwork Conference	Singapore Nov 2021
APSEC 2020 <i>Program Co-Chair</i> The 26th Asia-Pacific Software Engineering Conference	Singapore Dec 2020
ICFEM 2020 <i>General Co-Chair</i> The 22th International Conference On Formal Engineering Methods	Singapore Nov 2020
NBAI 2019 <i>General Co-Chair</i> NTU – BGU 1st Joint Workshop on AI SECURITY	Singapore 18 Oct 2019
MLST 2019 <i>General Co-Chair</i> The International Workshop on Machine Learning and Software Testing	Xi'an, China 23 April 2019
AI4Mobile 2019 <i>General Co-Chair</i> The first International Workshop on Artificial Intelligence for Mobile	Hangzhou, China 24 Feb 2019
ICFEM 2018 <i>Workshop Chair</i> The 20th International Conference On Formal Engineering Methods	Australia Nov 2018
SG-CRC 2018 <i>Program Co-Chair</i> The 3rd Singapore Cyber Security R&D Conference	Singapore Mar 2018
UMAP 2018 <i>Financial Chair</i> User Modelling, Adaptation and Personalization	Singapore July 2018
SG-CRC 2017 <i>Program Co-Chair</i> The 2nd Singapore Cyber Security R&D Conference	Singapore Feb 2017
ESSS 2016 <i>General Co-Chair</i> The Fifth International Workshop on Engineering Safety and Security Systems	Cyprus Nov 2016
ASE 2016 <i>Tool Demo Chair</i> 31th IEEE/ACM International Conference on Automated Software Engineering	Singapore Nov 2016
ICECCS 2015 <i>General Chair</i>	Australia Dec 2015

The 21th International Conference on Engineering of Complex Computer Systems	
ESSS 2015	Oslo, Norway
<i>General Co-Chair</i>	<i>May 2015</i>
The Four International Workshop on Engineering Safety and Security Systems	
PRDC 2014	Singapore
<i>Program Co-Chair</i>	<i>Nov 2014</i>
The 20th IEEE Pacific Rim International Symposium on Dependable Computing	
ICFEM 2014	Luxembourg
<i>Publicity Chair</i>	<i>Nov 2014</i>
The 16th International Conference On Formal Engineering Methods	
ICECCS 2014	Tianjin China
<i>Publicity Chair</i>	<i>Aug 2014</i>
The 19th International Conference on Engineering of Complex Computer Systems	
IFIPTM 2014	Singapore
<i>Local Chair</i>	<i>July 2014</i>
The 8th IFIP WG 11.11 International Conference on Trust Management	
FSFMA 2014	Singapore
<i>General Chair</i>	<i>May 2014</i>
The 2nd French Singaporean Workshop in Formal Methods and Applications	
ESSS 2014	Singapore
<i>Program Co-Chair</i>	<i>May 2014</i>
The Third International Workshop on Engineering Safety and Security Systems	
FM 2014	Singapore
<i>Financial Chair and Local Chair</i>	<i>May 2014</i>
The 19th International Symposium on Formal Methods	
ACOMP 2013	Ho Chi Minh City, Vietnam
<i>Track Co-Chair</i>	<i>October 23-25, 2013</i>
The International Conference on Advanced Computing and Applications on Advances in Software Engineering Track	
FSFMA 2013	Singapore
<i>Organizing Committee</i>	<i>July 2013</i>
The 1st French Singaporean Workshop in Formal Methods and Applications	
ACCESS 2013	Singapore
<i>Program Co-Chair</i>	<i>2013</i>
The International Congress on Advances in Citizen, Cyberspace & Environment Safety and Security	
ESSS 2013	Luxembourg
<i>Program Co-Chair</i>	<i>March 2013</i>
The Second International Workshop on Engineering Safety and Security Systems	
ICECCS 2013	Singapore
<i>Program Co-Chair</i>	<i>July 2013</i>
The 18 IEEE International Conference on Engineering of Complex Computer Systems	
ICFEM 2013	New Zealand
<i>Workshop and Tutorial Chair</i>	<i>Nov 2013</i>
The 13th International Conference on Formal Engineering Methods	
ESSS 2012	Singapore
<i>Program Co-Chair</i>	<i>Feb 2012</i>
The First International Workshop on Engineering Safety and Security Systems, Singapore	

MoCSeRS 2010*Program Co-Chair*

The 1st International Workshop on Model Checking Secure and Reliable Systems

Singapore*June 2010***SSIRI 2010***Registration Chair*

The 4th IEEE International Conference on Secure Software Integration and Reliability Improvement

Singapore*June 2010***ATVA 2010***Local Chair*

The 8th International Symposium on Automated Technology for Verification and Analysis

Singapore*September 2010***(4) Program Committee Member**.....

1. The 32nd ACM Conference on Computer and Communications Security (CCS 2025), Taiwan, 2025
2. The 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025), Korea, 2025
3. The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2025), Norway, 2025
4. The 47th IEEE/ACM International Conference on Software Engineering (ICSE 2025), Canada, 2025
5. The 39th IEEE/ACM International Conference on Automated Software Engineering (ASE 2024), United State, 2024
6. The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2024), Austria, 2024
7. The 46th International Conference on Software Engineering (ICSE 2024), Portugal, 2024
8. First International Workshop on Large Language Models for Code (LLM4Code 2024), Portugal, 2024
9. The 19th ACM ASIA Conference on Computer and Communications Security (ACM AsiaCCS 2024), Singapore, 2024
10. The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering Industrial Track (Industrial Track@FSE 2022), Singapore, 2022.
11. The 34th International Conference on Computer Aided Verification (CAV 2022), Haifa, Israel. August 7-10, 2022.
12. The 37th IEEE/ACM International Conference on Automated Software Engineering (ASE 2022), USA, 2022.
13. The 59th Design Automation Conference (DAC 2022), July, USA, 2022.
14. Mining Software Repositories (MSR 2022), Pittsburgh, Pennsylvania, USA, 2022.
15. The 14th Internetwork Conference, China, 2022.

16. The 38th IEEE International Conference on Software Maintenance and Evolution (ICSME 2022), Cyprus, 2022.
17. The 44th International Conference on Software Engineering (ICSE 2022) DS - Doctoral Symposium, USA, 2022.
18. The 58th Design Automation Conference (DAC 2021), July, USA, 2021.
19. Mining Software Repositories (MSR 2021), Madrid, Spain, May 2021.
20. The 37th IEEE International Conference on Software Maintenance and Evolution (ICSME 2021), 2021.
21. The 27th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2021), Nov, 2021.
22. The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2021), Aarhus, Denmark, 2021.
23. The 36th IEEE/ACM International Conference on Automated Software Engineering (ASE 2021), Melbourne, Australia, 2021.
24. The IEEE/ACM 43rd International Conference on Software Engineering (ICSE 2021), May 20-30, 2021. Madrid, Spain, 2021.
25. The 32nd International Conference on Computer-Aided Verification (CAV 2020), July 19-24, 2020 in Los Angeles, California, USA.
26. The the 57th Design Automation Conference (DAC 2020), San Francisco, CA, July 19-23, USA, 2020.
27. The 35th IEEE/ACM International Conference on Automated Software Engineering (ASE 2020), Mon 21 - Fri 25 September 2020 Melbourne, Australia.
28. The 15th ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2020), Taipei, Taiwan, 1-4 Jun 2020.
29. Mining Software Repositories (MSR 2019), Montreal, QC, Canada, Sun 26 - Mon 27, May 2019.
30. The 18th International Conference on Software and Systems Reuse (ICSR 2019), Cincinnati, OH, USA, June 26-28, 2019.
31. The 13th International Symposium on Theoretical Aspects of Software Engineering (TASE 2019), Guilin, China, July 29-31, 2019.
32. the Twenty-first International Conference on Formal Engineering Methods (ICFEM 2019), Shenzhen, China, November 5th-9th, 2019.
33. The 13th IEEE International Symposium on Service-Oriented System Engineering (SOSE 2019), California, USA, April, 4 - 9, 2019.
34. The 34th IEEE/ACM International Conference on Automated Software Engineering (ASE 2019), USA, September 3 to 7, 2019.

35. The 26th ACM Conference on Computer and Communications Security (CCS 2019), Toronto, Canada, October 15 - 19, 2019.
36. 23rd International Symposium on Formal Methods - 3rd World Congress on Formal Methods (FM 2019), Porto, Portugal, October 7-11, 2019.
37. The 27th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (FSE 2019), Tallinn, Estonia, 26-30 August 2019.
38. The 25th ACM Conference on Computer and Communications Security (CCS 2018), Toronto, Canada, October 15 - 19, 2018.
39. The 16th International Symposium on Automated Technology for Verification and Analysis (ATVA 2018), Los Angeles, California, USA, October 7-10, 2018.
40. The 33rd IEEE/ACM International Conference on Automated Software Engineering (ASE 2018), Corum, Montpellier, France from September 3 to 7, 2018.
41. The 34th IEEE International Conference on Software Maintenance and Evolution (ICSME 2018), Madrid, Spain, September 23 to 29, 2018.
42. The 24th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2018), Nov, 2018.
43. The 12th IFIP WG 11.11 International Conference on Trust Management (IFIPTM 2018), 2018.
44. The 11th IEEE Conference on Software Testing, Validation and Verification (ICST 2018), 9 Apr - 13 Apr 2018.
45. The 12th International Symposium on Service-Oriented System Engineering (SOSE 2018), June 2018.
46. The 23rd International Conference on Engineering of Complex Computer Systems (ICECCS 2018), Melbourne, Australia, 12-14 December 2018.
47. The 26th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) - Formal Demonstration Track (FSE 2018 Demo), Florida, United States, Sun 4 - Fri 9 November 2018.
48. The 27th International Symposium on Software Testing and Analysis (ISSTA 2018), Amsterdam, Netherlands, 2018.
49. The 24th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2018), Thessaloniki, Greece, April, 2018.
50. Sixth International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS 2018), Gold Coast, Australia, November 16, 2018.
51. 34th International Conference on Software Maintenance and Evolution (ICSME), Madrid, Spain, Sep 2018.
52. the 39th International Conference on Software Engineering (ICSE 2017) Student Research Competition, May, 2017.

53. The 23rd IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2017), Nov, 2017.
54. The 11th International Symposium on Service-Oriented System Engineering (SOSE 2017), June 2017.
55. The 19th International Conference On Formal Engineering Methods (ICFEM 2017). November 2017.
56. The 22th International Conference on Engineering of Complex Computer Systems (ICECCS 2017), Fukuoka, Japan, November 6-8, 2017.
57. The 10th IEEE Conference on Software Testing, Validation and Verification (ICST 2017), Tokyo, 13-17 March, 2017.
58. The 32rd IEEE/ACM International Conference on Automated Software Engineering (ASE 2017), 2017.
59. The 18th International Conference On Formal Engineering Methods (ICFEM 2016). Tokyo, Japan 14-18 November 2016.
60. the IEEE International Conference on Software Testing, Verification and Validation: Testing Tool Demos Track (ICST 2016). Chicago, IL, USA April 10-15, 2016.
61. The 19th IEEE International Symposium on Object/component/service-oriented Real-time distributed Computing (ISORC 2016). York, UK, May 2016.
62. The 10th International Symposium on Service-Oriented System Engineering (SOSE 2016), Kongsberg, Norway, June 2016.
63. The 25th International Joint Conference on Artificial Intelligence (IJCAI 2016), New York, USA, July, 2016.
64. The 31th IEEE/ACM International Conference on Automated Software Engineering (ASE 2016), Singapore Nov 2016.
65. Doctoral Symposium of The 17th International Conference On Formal Engineering Methods (FM Doc Sym 2015). Oslo, Norway, June 22-26, 2015.
66. The 17th International Conference On Formal Engineering Methods (ICFEM 2015). Paris, France, Nov, 2015.
67. The 9th IFIP WG 11.11 International Conference on Trust Management (IFIPTM 2015), 15 - 29 July, Hamburg, Germany, 2015
68. The 18th IEEE International Symposium on Object/component/service-oriented Real-time distributed Computing (ISORC 2015). Auckland, New Zealand, April 13 - 17, 2015.
69. The 21th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2015), 18 - 20 Nov, Zhangjiajie, China, 2015
70. The 9th International Symposium on Service-Oriented System Engineering (SOSE 2015), March 30 - April 3, San Francisco Bay, USA, 2015.

71. The 24th IEEE International Conference on Enabling Technologies: Infrastructure for Collaborative Enterprises (WETICE 2015), Larnaca, Cyprus, 2015
72. The 10th International Workshop on Security and Trust Management (STM 2014). Wroclaw, Poland, September 10-11, 2014.
73. The 21st Asia-Pacific Software Engineering Conference (APSEC 2014), Jeju Island, Korea, 1-4 December 2014.
74. The 12th IEEE International Conference on Pervasive Intelligence and Computing (PCom 2014), Dalian, China, August 24-27, 2014.
75. The 3rd International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS 2014). Luxembourg, 2014.
76. The 8th IFIP WG 11.11 International Conference on Trust Management (IFIPTM 2014), 7 - 10 July, Singapore, 2014
77. The 16th International Conference On Formal Engineering Methods (ICFEM 2014), 3 - 7 Nov, Luxembourg, 2014
78. The 20th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2014), Singapore, 2014
79. The 19th International Conference on Engineering of Complex Computer Systems (ICECCS 2014), 4 - 7 Aug, Tianjin China, 2014
80. The Seventh International Conference on Dependability (DEPEND 2014) November 16 - 20, 2014 - Lisbon, Portugal
81. The 8th International Symposium on Service-Oriented System Engineering (SOSE 2014), Oxford, UK. April 7-11 2014.
82. The 17th IEEE International Symposium on Object/component/service-oriented Real-time distributed Computing (ISORC 2014). Reno, Nevada, U.S.A., June 10-12, 2014.
83. The 29th Symposium On Applied Computing, Track on Software Verification and Testing (SAC-SVT 2014). Gyeongju, Korea, March 24 - 28, 2014.
84. The 2nd International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS 2013). New Zealand. Nov 2013.
85. The 20th Asia-Pacific Software Engineering Conference (APSEC 2013). Bangkok, Thailand. December 2013.
86. The 13th International Conference on Formal Engineering Methods (ICFEM 2013). New Zealand. Nov 2013.
87. The 19th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2013). Vancouver, British Columbia, Canada, December 2-4, 2013.
88. The 16th IEEE International Symposium on Object/component/service-oriented Real-time distributed computing (ISORC 2013). Paderborn Germany, June 19-21, 2013.

89. The 28th Symposium On Applied Computing, Track on Software Verification and Testing (SAC-SVT 2013). Coimbra, Portugal, March 18th to 22nd, 2013.
90. The 1st International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS 2012). Kyoto, Japan, November 12, 2012.
91. The 12th International Conference on Formal Engineering Methods (ICFEM 2012). Kyoto, Japan, November 12, 2012.
92. The 19th Asia-Pacific Software Engineering Conference (APSEC 2012). December 4 - 7, Hong Kong, 2012.
93. The 2012 Summer FTRA International Symposium on Advances in Cryptography, Security and Applications for Future Computing (ACSA-Summer 2012). Vancouver, Canada, June 26-28, 2012.
94. The 50th International Conference on Objects, Models, Components and Patterns (TOOLS Europe 2012).
95. The 36th Annual Computer Software and Applications Conference (COMPSAC 2012).
96. The 15th IEEE International Symposium on Object/component/service-oriented Real-time distributed computing (ISORC 2012).
97. The Sixth IEEE International Symposium on Theoretical Aspects of Software Engineering (TASE 2012). Beijing, China. 4-6 July 2012.
98. R50: Rediscovering 50 Years of Discoveries workshop, associated with the 50th annual meeting of the Association for Computational Linguistics (ACL 2012), Jeju Island Korea. July 8-14, 2012
99. The 4th International Workshop on Testing, Analysis and Verification of Web Software (TAV-WEB 2010).

(5) Research Funding (50M+ SGD External Grant as PI)

Desay: 336,000 SGD	Singapore
<i>Principal Investigator</i>	2026 - 2026
Automated Open-Source Component Analysis (AutoSCA) and Emulation Based Fuzzing Tool (EmuFuzz) for Automotive Software	
Desay: 222,100 SGD	Singapore
<i>Principal Investigator</i>	2026 - 2026
Multi-Agent Framework for Requirements Extraction and Development	
DSTA: 3,601,000 SGD	Singapore
<i>Principal Investigator</i>	2026 - 2028
Fuzz Testing and Traffic Analysis for Drone Protocol	
NRF CREATE: 20,000,000 SGD	Singapore
<i>Principal Investigator</i>	2024 - 2027
Engineering the Security of Future Medical Devices and Systems	
RIPPLE: 350,000 USD	Singapore
<i>Principal Investigator</i>	15 July 2024 - 14 July 2026
Autonomous AI Agent Network on Blockchain	

DTC Research Grant Call: 358,800 SGD	Singapore
<i>Co-Principal Investigator</i>	15 April 2024 - 14 April 2027
Combatting Prejudice in AI: A Responsible AI Framework for Continual Fairness Testing, Repair, and Transfer	
CSA: 1,500,000 SGD	Singapore
<i>Co-Principal Investigator</i>	4 Mar 2024 - 7 Jan 2025
Smart Nation IoT-system Threat Analytics	
MOE Tier 1: 100,000 SGD	Singapore
<i>Principal Investigator</i>	1 Mar 2024 - 31 Aug 2025
Holistic and Practical Remediation of Software Vulnerability	
IGS: 1,100,000 SGD	Singapore
<i>Principal Investigator</i>	15 Dec 2023 - 14 Dec 2027
Theme 3: Algorithms for Privacy, Security and Provenance	
CoE CRP and Tier 3 Preparatory Grant Call 2023: 50,000 SGD	Singapore
<i>Principal Investigator</i>	1 Nov 2023 - 31 Oct 2024
Building H3 (Helpful, Honest, and Harmless) Large Scale Models	
CSA: 61,911,200 SGD	Singapore
<i>Co-Principal Investigator</i>	1 Aug 2023 - 31 July 2027
CyberSG R&D Programme Office (CRPO)	
AISG: 1,510,800 SGD	Singapore
<i>Principal Investigator</i>	1 July 2023 - 30 June 2025
Towards Building Unified Autonomous Vehicle Scene Representation for Physical AV Adversarial Attacks and Visual Robustness Enhancement	
Desay: 485,000.00 SGD	Singapore
<i>Principal Investigator</i>	30 June 2023 - 29 Dec 2024
Enhancing the Robustness of Object Detection against Natural Corruption-aware Adversarial Attacks	
Desay: 519,000.00 SGD	Singapore
<i>Principal Investigator</i>	30 June 2023 - 29 Dec 2024
Simulation-based SOTIF (Safety of the Intended Functionality) Testing for Autonomous Driving Systems	
AISG: 280,200 SGD	Singapore
<i>Principal Investigator</i>	8 Sep 2023 - 7 March 2025
MAASS: Malware Authorship Attribution based on Source Code and Static Analysis	
AISG: 1,764,000 SGD	Singapore
<i>Co-Principal Investigator</i>	1 Nov 2023 - 30 April 2025
Towards Semantic-Aware Multimodal and Multilingual Deep Learning Systems for E-Commerce Applications	
Desay: 662,000.00 SGD	Singapore
<i>Principal Investigator</i>	1 May 2022 - 30 April 2024
Emulation Based Fuzzing Technique	
MOE: 524,700 SGD	Singapore
<i>Co-Principal Investigator</i>	14 Feb 2022 - 13 Feb 2025
A Framework for Intellectual Property Protection of Deep Learning Applications	
CSA: 469,200.00 SGD	Singapore
<i>Principal Investigator</i>	16 Nov 2022 - 15 Nov 2024
CloudSec: Security Analysis of Cloud Infrastructures	
CSA: 12,364,100 SGD	Singapore
<i>Principal Investigator</i>	1 Aug 2022 - 31 July 2026
Trustworthy AI Centre NTU (TAICeN)	

NTU: 5,000 SGD <i>Principal Investigator</i> CoE Innovator (Entrepreneurship) Award 2022	Singapore 1 March 2022 - 31 Aug 2023
CSA: 4,019,000 SGD <i>Co-Principal Investigator</i> Smart Nation Internet-of-Things System Threat Analytics	Singapore 5 March 2021 - 4 March 2024
AI Singapore: 6,882,200 SGD <i>Principal Investigator</i> TrustFUL: Trustworthy Federated Ubiquitous Learning	Singapore 1 April 2021 - 31 March 2025
NICE: 237,600 SGD <i>Principal Investigator</i> Security Risk Assessment of Android Apps	Singapore 1 April 2021 - 31 March 2022
MOE Tier 3 Grant: 9,990,675.6 SGD <i>Co-Principal Investigator</i> The Science of Certified AI Systems	Singapore 1 April 2021 - 31 March 2025
MOE Tier 2 Grant: 566,330 SGD <i>Principal Investigator</i> Smart Safe and Robust Motion Control for Multi-Robot Systems	Singapore 1 Feb 2021 - 31 Jan 2024
Desay: 550,000.00 SGD <i>Principal Investigator</i> Automatic Open Source Component Detection and Security Risk Management for Automotive Systems	Singapore 17 Aug 2020 - 16 Feb 2022
NRF: 2,503,359.60 SGD <i>Principal Investigator</i> Towards Building Trustworthy and Robust Intelligent Systems and its Application on Human Brain	Singapore 1 Feb 2020 - 31 Jan 2024
Continental Corp Lab: 4,666,475 SGD <i>Principal Investigator</i> Automotive Software Security Analysis	Singapore 1 Dec 2019 - 30 Nov 2024
Continental Corp Lab: 1,537,747 SGD <i>Co-Principal Investigator</i> AI for SW Engineering	Singapore 1 Dec 2019 - 30 Nov 2024
Nanyang Award of Research Excellence (Y-INV): 15,000 SGD <i>Principal Investigator</i> An Empirical Evaluation of GDPR Compliance Violations in Mobile Apps and IoT Devices	Singapore 1 Oct 2019 - 30 Sep 2020
NRF: 3,849,762.60 SGD <i>Principal Investigator</i> Enhancing Cyber Resilience of Deep Learning Models Against Adversarial Cyber Attacks	Singapore 1 April 2019 - 31 March 2022
NTU: 50,000 SGD <i>Principal Investigator</i> 2019 University Leadership Forum Chair in Computer Science and Engineering	Singapore 1 April 2019 - 31 March 2023
AISG: 4,221,600.00 SGD <i>Co-Principal Investigator</i> An end-to-end Adaptive AI-Assisted 3H Care (A3C) System	Singapore 1 Jan 2019 - 31 Dec 2021
NRF: 1,283,475 SGD <i>Principal Investigator</i> National Satellite of Excellence of Singapore in Trustworthy Software Systems	Singapore 1 Jan 2019 - 31 Dec 2022

HP Corp Lab: 3,106,744 SGD <i>Principal Investigator</i> End to End Attestation Infrastructure	Singapore 5 Oct 2018 - 4 Oct 2022
HP Corp Lab: 3,166,744 SGD <i>Principal Investigator</i> Methodologies and Techniques to Evaluate Malware Evolution and Mitigations	Singapore 5 Oct 2018 - 4 Oct 2022
HP Corp Lab: 3,281,964 SGD <i>Principal Investigator</i> Secure Machine Learning	Singapore 5 Oct 2018 - 4 Oct 2022
SCSE: 43,400 SGD <i>Principal Investigator</i> Distributed and Automated Mobile App Testing Platform	Singapore 1 Dec 2017 - 30 Jun 2018
NGF: 242,500 SGD <i>Principal Investigator</i> Cloud-based Mobile App Testing Service	Singapore 1 Jan 2018 - 30 Jun 2019
NRF: 1,182,000 SGD <i>Principal Investigator</i> Smart Binary-level Vulnerability Assessment for Cyber-attack Prevention	Singapore 1 Oct 2017 - 30 Sep 2019
US Army: 74,000 USD <i>Principal Investigator</i> Scalable Malware and Vulnerability Analysis Using Program Metrics	USA 25 Sep 2017 - 24 Sep 2018
NRF: 90,000 SGD <i>Principal Investigator</i> Mobile (iOS) Security Study for Cyber-Attack Prevention	Singapore 1 Nov 2017 - 31 Oct 2018
DSO: 552,000 SGD <i>Principal Investigator</i> BINSEC: Binary Analysis for Security	Singapore 1 Apr 2017 - 31 Mar 2020
MOE: 242,580 SGD <i>Co-Principal Investigator</i> Productive Failure via Educational Games for Tertiary Education	Singapore 1 Jan 2017 - 31 Dec 2019
NRF: 399,600 SGD <i>Co-Principal Investigator</i> Improving Cybersecurity through Optimal Policy Design and Human Behaviour Modelling	Singapore 1 Jan 2017 - 31 Dec 2019
FSTD: 516,500 SGD <i>Principal Investigator</i> ACRONIS: Cybersecurity for Autonomous Vehicles	Singapore 1 Nov 2016 - 31 Oct 2019
MoE Tier 2 Grant: 659,078 SGD <i>Principal Investigator</i> Robust Control of Large Scale Concurrent Systems with Unreliable Resources	Singapore 1 March 2016 - 28 Feb 2019
Wincor: 60,000 SGD <i>Principal Investigator</i> Security Enhancements For ATM And POS Systems	Singapore 21 Sep 2015 - 31 Dec 2019
DSO: 69,000 SGD <i>Principal Investigator</i> Software Vulnerability Discovery Tool Building	Singapore 1 May 2015 - 30 April 2016
NRF-Cybersecurity: 2,114,565 SGD <i>Principal Investigator for Sub-Project 3</i>	Singapore 1 Jan 2015 - 31 Dec 2019

Securify: A Compositional Approach of Building Security Verified System	
NRF-Cybersecurity: 2,032,913 SGD	Singapore
<i>Co-Principal Investigator for Sub-Project 2</i>	<i>1 Jan 2015 - 31 Dec 2019</i>
Securify: A Compositional Approach of Building Security Verified System	
NRF-Cybersecurity: 850,000 SGD	Singapore
<i>Co-Principal Investigator for Sub-Project 1</i>	<i>1 Jan 2015 - 31 Dec 2019</i>
Securify: A Compositional Approach of Building Security Verified System	
MoE Tier 1: 100,000 SGD	Singapore
<i>Principal Investigator</i>	<i>1 Nov 2014 - 31 Oct 2017</i>
Distributed Plant Modelling, Fault diagnosis, and Supervisor Control of Large Scale Automated Manufacturing Systems	
EDEX GRANT: 31,000 SGD	Singapore
<i>Principal Investigator</i>	<i>10 Dec 2014 - 31 June 2016</i>
Gamifying and Socializing Teaching and Learning of Software Engineering	
CSIT: 471,000 SGD	Singapore
<i>Principal Investigator</i>	<i>1 April 2014 - 31 March 2017</i>
Vulnerability Detection in Binary Code	
MoE Tier 1: 50,000 SGD	Singapore
<i>Principal Investigator</i>	<i>March 2014 - Feb 2016</i>
Bring the Advanced Model Checking Techniques to Real-world Problems	
ROLLS-ROYCE@NTU: 621,117 SGD	Singapore
<i>Co-Principal Investigator (35% Cash)</i>	<i>1 Aug 2013 - 31 July 2017</i>
Business Process Simulation Description Language (BPSDL)	
NTU-NAP: S\$900,000	Singapore
<i>Principal Investigator</i>	<i>1 May 2013 - 30 April 2016</i>
Formal Verification on Cloud	
STIC-Asia Funding: 30,000 Euro	France, Singapore and Vietnam
<i>Co-Principal Investigator</i>	<i>2013 - 2014</i>
CATS - Compositional Analysis of Timed Systems	
STIC is Information and Communication Science and Technologies. The program's objective is to promote the internationalization of French research, strengthen high-level collaborations with Asian emerging and developed on the basis of mutual benefit, and contribute to the regional integration of the area through the promotion of research excellence and community support Asian scientists in all fields of science and information technology and communication (ICT).	
NTU-SUG: S\$100,000	Singapore
<i>Principal Investigator</i>	<i>1 Oct 2012 - 31 Sep 2014</i>
Automatic Checking and Verification of Security Protocol Implementations. Account No. M4080996.020.500000	
Merlion Funding by French Embassy of Singapore: S\$60,000	Singapore
<i>Co-Principal Investigator</i>	<i>1 March 2012 - 1 March 2014</i>
Software Verification from Design to Implementation	
DRTech-Temasek Research Fellow (TRF) Funding: S\$1,150,000	Singapore
<i>Principal Investigator</i>	<i>6 Oct 2011 - 5 Oct 2014</i>
Research and Development in the Formal Verification of System Design and Implementation. WBS No. R394-000-063-23	

(6) Invited Talks.....

1. *AI and Devsecops*, Huawei STW2023, Sep 2023.

2. *AISG and Cybersecurity*, IMDA Technical Sharing Session #3 on Trustworthy AI, Oct 2023, Singapore.
3. *The research and innovation of NTU and CRPO Incubator*, Chaoyang District - Singapore Internet 3.0 Industry Symposium, Oct 2023, Singapore.
4. *AISG and Cybersecurity*, ST Eng CTO-Gen AI Round Table with IHLs, Nov 2023, Singapore
5. *The Software Genome Project*, ICG-18-II, Dec 2023, Shenzhen
6. *The Software Genome Project*, Software Security Technology Forum and Third Anniversary Release Event, Dec 2023.
7. *AI+Security+Web3*, The First China-Europe Nanyang Forum, Feb 2024, Singapore.
8. *Intelligent Security Testing and Analysis of Automotive Systems*, Tencent, March, 2022, China.
9. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, Beihang University, Dec, 2021, China.
10. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, China Desay, Dec, 2021, Huizhou, China.
11. *A Road Towards of Software Testing*, Huawei, Nov, 2021, China.
12. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, Beihang University, Nov, 2021, China.
13. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, Nanjing University, Sep, 2021, China.
14. *The Journey of Fuzzing Research*, Nanjing University, June, 2021, China.
15. *The Journey of Fuzzing Research*, Ohio State University, June, 2021, USA.
16. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, Nanjing University of Aeronautics and Astronautics, June, 2021.
17. *The Journey of Fuzzing Research*, InforSec, Jan, 2021, China.
18. *Intelligent Security Testing and Analysis of Automotive Systems*, China Desay, Dec, 2020, Huizhou, China.
19. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, China Unicom, November, 2020, Guangzhou, China.
20. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, Harbin Institute of Technology, November, 2020, Harbin, China.
21. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, Institute of Information Engineering, Chinese Academy of Sciences (IIE, CAS), October, 2020, Beijing, China.
22. *The Journey of Fuzzing Research*, Ohio State University (OSU), October, 2020, USA.

23. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, University of Science and Technology of China, October, 2020, Hefei, China.
24. *Security of Intelligent Automotive Systems*, Continental AG, September, 2020, Singapore.
25. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, Nanjing University of Aeronautics and Astronautics (NUAA), September, 2020, Nanjing, China.
26. *The Journey of Taming Vulnerabilities: from programs, libraries to software supply chain*, Shenzhen University, September, 2020, Shenzhen, China.
27. *The Journey of Fuzzing Research*, Summer School at Zhejiang University, August, 2020, Hangzhou, China.
28. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, Wuhan University, May, 2020, Wuhan, China.
29. *Security Testing - The journey to hunt vulnerabilities*. Invited Talk in Nanjing University, 20 Dec 2019, China.
30. *Security Testing - Our Attempts and Lesson Learned*. Invited Talk in National Software Testing Competition, 1st Dec 2019, China.
31. *Software Supply Chain Security*. Invited Talk in NASAC 2019- Software Security Industrial Forum, 24 Nov 2019, China.
32. *Data Driven Software Intelligent Engineering*. Invited Talk in NASAC 2019- Data Driven SE Forum, 23 Nov 2019, China.
33. *Intelligent Software Engineering: Combining Deep Learning and Software Engineering*. Invited Talk in Continental Global Conference, 12 Nov 2019, Germany.
34. *NTU Cybersecurity Research*. Invited Talk in NTU – BGU 1st Joint Workshop on AI SECURITY, 17 October 2019, Singapore.
35. *NTU Cybersecurity Research*. Invited Talk in PVCS Workshop, 8 Aug 2019, Singapore.
36. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics, Artificial Intelligence to Blockchain*. Invited Talk in Byte Dance, 15 July 2019, Beijing, China.
37. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics, Artificial Intelligence to Blockchain*. Invited Talk in 3rd Cybersecurity Young Scientist Forum, 11 May 2019, Wuhan, China.
38. *AI + Cybersecurity*. Invited talk in First NTU “AI+X” Symposium on 2 and 3 May 2019, Singapore.
39. *Data Driven Vulnerability Analysis and Management*. Invited talk in ICST Industry Forum on Software Testing and Verification, 26 April 2019, Xi'an China.
40. *Secure the Vehicles with Offline and Online Techniques*. Invited talk in the 2nd Huawei Automotive Cybersecurity Workshop, 11 April 2019, Singapore.

41. *Data Driven Software Security Analysis*. Invited talk in the 26th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2019) Asia-Pacific Track, Feb 2019, China.
42. *Smart Contract Security Analysis for Fun and Profit*. Invited Talk in First Inter-College Blockchain Symposium, Jan 2019, Singapore.
43. *Secured Deep Learning Engineering*. Invited Talk in Interpol-NTU Workshop, Jan 2019, Singapore.
44. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics, Artificial Intelligence to Blockchain*. Invited Talk in Zhejiang University, Dec 2018, Hangzhou, China.
45. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics, Artificial Intelligence to Blockchain*. Invited Talk in Ant Finance, Dec 2018, Hangzhou, China.
46. *Data Driven Software Engineering and Software Engineering of Data Driven Programming*. Invited Talk in Data Driven Software Development Forum, Nov 2018, Shenzhen, China.
47. *Towards Building a Generic Vulnerability Detection Platform*. Invited Talk in the 7th Asian-Pacific Workshop of Advanced Software Engineering, Nov 2018, Australia.
48. *Secured Deep Learning Engineering*. Invited Talk in Yanqi-Lake Meeting on Software Automation, Nov 2018, Beijing, China.
49. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics, Artificial Intelligence to AV*. Invited Talk in WASP-NTU Workshop, Oct 2018, Singapore.
50. *Secured Machine Learning Engineering*. Invited Talk in Industry-NTU Dialogue, 21 Sep 2018, Singapore.
51. *Introduction of NTU Cybersecurity: from Formal Methods, Program Analysis, Data Analytics, Artificial Intelligence to Block Chain*. Invited Talk in IMDA-NTU Workshop, Sep 2018, Singapore.
52. *AI vs. Testing*. Invited Talk in Intelligent Software Engineering Forum, Sep 2018, Nanjing, China.
53. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics to Artificial Intelligence*. Invited Talk in KAIST, 27 Aug 2018, South Korea.
54. *Introduction of NTU Cybersecurity: from Formal Methods, Program Analysis, Data Analytics to Artificial Intelligence*. Invited Talk in National Security Engineering Centre (NSEC), Aug 2018, Singapore.
55. *Cyber-security: the Journey from Formal Methods, Program Analysis to Data Analytics*. Invited Talk in Singapore Cybersecurity Lean LaunchPad, June 2018, Singapore.
56. *Cyber-security: the Journey from Formal Methods, Program Analysis to Data Analytics*. Invited Talk in Hangzhou International Conference on Frontiers of Data Science, May 2018, China.
57. *Cyber-security: the Journey from Formal Methods, Program Analysis to Data Analytics*. Invited Talk in Huawei Singapore, April 2018, Singapore.
58. *AI for Security and AI's Security*. Invited Talk in NTU AI Security Workshop, April 2018, Singapore.

59. *Introduction of NTU Cybersecurity: from Formal Methods, Program Analysis to Data Analytics*. Invited Talk in Qihoo 360, March 2018, China.
60. *Securify: A Compositional Approach of Building Security Verified System*. The 3rd Singapore Cyber Security R&D Conference, March 2018, Singapore.
61. *Cyber-security Research at NTU*. Invited Talk in HP-NTU Workshop, Jan 2018, Singapore.
62. *The Unknown Journey of Academics*. Invited Talk in NUS Ph.D. Training Workshop, Jan 2018, Singapore.
63. *Building Effective Binary Fuzzers for all Architectures and Platforms*. Invited Talk in Huawei, Dec 2017, Shenzhen, China.
64. *Intelligent (Binary Program) Vulnerability Analysis and Management*. Invited Talk in Desay-NTU Cybersecurity Workshop, Nov 2017, Singapore.
65. *Intelligent Cyber-Security Analysis with the Help of Big Data*. Invited Talk in HKU-NTU Forum, Nov 2017, Singapore.
66. *Intelligent Cyber-Security Analysis with the Help of Big Data*. Invited Talk in NASAC 2017, Nov 2017, China.
67. *The Influence of Cybersecurity in Society*. Invited Talk in the Future Leaders Program of STS Forum, Oct 2017, Japan.
68. *Can We Test Better? - Our Attempts and Lesson Learned*. Invited Talk in East China Normal University, Oct 2017, Shanghai, China.
69. *NTU Cybersecurity Research*. Invited Talk in CSA-NTU Workshop, Sep 2017, Singapore.
70. *NTU Cybersecurity Research*. Invited Talk in XCTF Competition Workshop, Aug 2017, Singapore.
71. *Cyber-security: the Journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in Sydney University, July 2017, Australia.
72. *Cyber-security: the journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in Griffith University, 27 June 2017, Australia.
73. *Cyber-security: the journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in InforSec with Live broadcasting, 6 June 2017, Beijing, China.
74. *Emergence of AI in Cybersecurity*. Invited Talk in NVidia Workshop on Cybersecurity, June 2017, Singapore.
75. *Cyber-security: the Journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in Alibaba Cloud Security Team, April 2017, China.
76. *Smart Binary-Level Vulnerability Assessment for Cyber-Attack Prevention*. Invited Talk in AIA, March 2017, Hongkong, China.
77. *Complete Security Verification*. Invited Talk in the 2nd Singapore Cyber Security R&D Conference, Feb 2017, Singapore.

78. *Building Trusted and Resilient Systems (BuTRS)*. Invited Talk in The 18th IEEE International Symposium on High Assurance Systems Engineering (HASE 2017), Jan 2017, Singapore.
79. *Cyber-security: the journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in HKUST - Hong Kong University of Science and Technology, Jan 2017, China.
80. *Cyber-security: the journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in Tsinghua University, Dec, 2016, China.
81. *Vulnerability Analysis for Binary Programs*. Invited Talk in 6th International Symposium on High Confidence Software (ISHCS 2016), Dec 2016, China.
82. *Cyber-security: the journey from Formal Methods, Program Analysis to and Data Analytics*. Invited Talk in Tian Jin University, Dec, 2016, China.
83. *Cyber-Security: The Journey From Formal Methods, Program Analysis To And Data Analytics*. Invited Talk in Rafael Workshop, Sep, 2016, Israel.
84. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in CSA Workshop, July 2016, Singapore.
85. *A Verification Framework for Stateful Security Protocols*. Invited Talk in ETH, July 2016, Switzerland.
86. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in University of Passau, June 2016, Germany.
87. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in Tel Aviv University, June, 2016, Israel.
88. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in Nanjing University, June, 2016, China.
89. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in National Institute of Advanced Industrial Science (AIST), 30 May 2016, Osaka, Japan.
90. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in Ben-Gurion University workshop, Feb 2016, Singapore.
91. *Securify: A Compositional Approach of Building Security Verified System*. Invited Talk in The first Singapore Cyber Security R&D Conference, Jan 2016.
92. *Cyber-Security: Formal Methods Approaches And Beyond*. Invited Talk in University of Technology, 14 Dec, 2015, Sydney Australia
93. *Cyber-Security: Formal Methods Approaches And Beyond*. The 21th International Conference on Engineering of Complex Computer Systems, Australia, Nov 2015
94. *Automatic Software Specification Learning and Summarization*. Invited Talk in Peking University, Oct 2015, Peking, China.
95. *Towards a Software Engineering Thinking of System Security*. Invited Talk in DSO, Aug 2015, Singapore.

96. *Applications of Model Checking on Different Systems using PAT*. Invited Talk in AFRL-Singapore Defence Workshop, May 2015, Singapore.
97. *Applying Formal Methods in Cyber-security Systems*. Invited Talk in National Institute of Advanced Industrial Science (AIST), 20 March 2015, Tsukuba, Japan.
98. *Applications of Model Checking on Different Systems using PAT*. Invited Talk in Hosei University, 19 March 2015, Tokyo, Japan.
99. *Malware and Vulnerability Detection using Static Analysis and Dynamic Analysis*. Invited Talk in Shonan Meeting, March 2015, Shonan, Japan.
100. *TrustFound: Towards a Formal Foundation for Model Checking Trusted Computing Platforms*. Invited Talk in SG-UK security workshop, 12 Feb 2015, Singapore.
101. *Applying Formal Methods in Cyber-security Systems*. Invited Talk in Shenzhen University, Dec 2014, China.
102. *Build Your Own Model Checker in One Month*. Tutorial in 35th International Conference on Software Engineering (ICSE 2013), 18th - 26th, 2013 San Francisco, USA.
103. *Pervasive Model Checking: An introduction to PAT*. Invited Talk in Huawei Shanghai. 27 Sep 2013, Shanghai, China.
104. *Pervasive Model Checking: An introduction to PAT*. Invited Talk in Shanghai Jiao Tong University, 26 Sep 2013, Shanghai China.
105. *Pervasive Model Checking: An introduction to PAT*. Invited Talk in Paris 13 University, Jan 2013, Paris, France.

TEACHING

- (1) Awards**.....
- | | |
|---|-----------------------------------|
| MOE. 242,580 SGD | Singapore |
| <i>Co-Principal Investigator</i> | <i>1 Jan 2017 - 31 Dec 2019</i> |
| Productive Failure via Educational Games for Tertiary Education | |
| NTU Teaching Grant: EDEX GRANT. 31,000 SGD | Singapore |
| <i>Principal Investigator</i> | <i>10 Dec 2014 - 31 June 2016</i> |
| Gamifying and Socializing Teaching and Learning of Software Engineering | |
- (2) Text-books, book chapters, monograph, software**.....
- 2017 Book:** Yang Liu, Jun Sun and Jin Song Dong, *Model Checking and Its Applications*. Adaptation, Learning, and Optimization. Springer
- 2012 Book Chapter:** Jun Sun, Yang Liu and Jin Song Dong, *Timed Automata*. CRC Handbook of Finite State Based Models and Applications, Jiachun Wang (Editor)
- (3) Development of teaching materials**.....
- | | |
|--|--------------------------|
| Nanyang Technological University | Singapore |
| <i>CE/CZ4062 Computer Security (System Security), Course Materials</i> | <i>Aug 2017–Dec 2017</i> |

Nanyang Technological University	Singapore
<i>CE/CZ2006 Software Engineering, Course Materials</i>	<i>Jan 2018–May 2018</i>

Nanyang Technological University	Singapore
<i>CZ3003 Software System Analysis And Design, Course Materials</i>	<i>Jan 2013–May 2013</i>

National University of Singapore	Singapore
<i>CS4211 Advanced Software Engineering, Course Materials</i>	<i>2009 Fall/2011 Fall</i>

(4) Short courses.....

Five Day Course on Software Supply Chain Security	China
<i>East China Normal University</i>	<i>March 1-20 2019</i>

Three Day Summer School Course on Formal Verification	China
<i>East China Normal University</i>	<i>July 13-16 2019</i>

Half Day Course on Bitcoin and Blockchain	Singapore
	<i>April 20 2019</i>

Half Day Course on Bitcoin and Blockchain	Singapore
	<i>Sep 24 2018</i>

Half Day Tutorial: Build Your Own Model Checker with PAT	Singapore
<i>31st International Conference on Automated Software Engineering (ASE 2016)</i>	<i>Sep 3rd 2016</i>

One Day Tutorial: Introduction to PAT and its Applications	Singapore
<i>Training Course for Singapore Defence Researchers and Related Industries Engineers</i>	<i>Aug 21th 2013</i>

Half Day Tutorial: Build Your Own Model Checker in One Month	San Francisco, USA
<i>The 35th International Conference on Software Engineering</i>	<i>May 20th 2013</i>

One Week Course: Software Model Checking	Tian Jin University, China
<i>Summer School Course in Tian Jin University</i>	<i>July 2011</i>

Half Day Tutorial: Introduction to PAT	Lero, Limerick, Ireland
<i>The 17th International Symposium on Formal Methods (FM 2011)</i>	<i>June 20th, 2011</i>

Short Course on Automatic Verification of Real-time Systems	Singapore
<i>DSO National Laboratories</i>	<i>May 2011</i>

Short Course on Software Verification	China
<i>ZheJiang University</i>	<i>May 2011</i>

(5) Service as external examiner.....

Thesis: Instruction Coverage for Android App Testing and Tuning	Luxembourg
<i>PILGUN Aleksandr in University of Luxembourg</i>	<i>Nov 2020</i>

Thesis: Formal Verification Based Design Model Repair	New Zealand
<i>Chenghao Cai in University of Auckland</i>	<i>March 2020</i>

Thesis: Analysing and Testing Android Apps	Australia
<i>Yifei Zhang in the University of New South Wales</i>	<i>Nov 2018</i>

Thesis: Applications of Perturbation Analysis in Probabilistic Model Checking	Singapore
<i>Yamilet Llerena in National University of Singapore</i>	<i>June 2018</i>

Thesis: Precise Yet Scalable Program Analysis via Symbolic Execution	Singapore
<i>Rasool Maghareh in National University of Singapore</i>	<i>Nov 2016</i>

Service

(1) Services to the University	
NTU CRP Panel for Informatics and Mathematics	NTU
<i>Alternate Expert Member</i>	<i>Jan 2025 - Dec 2026</i>
School Search Committee for Tenure-Track Faculties	SCSE, Singapore
<i>Member</i>	<i>5 August 2021 - 31 Dec 2023</i>
NTU Research Council (NTURC)	NTU, Singapore
<i>Discipline Cluster Committee Member for EP2</i>	<i>20 Sep 2021 - 31 Dec 2023</i>
NRF-I Evaluation Panel	NTU, Singapore
<i>Member</i>	<i>March 2021</i>
CRP Evaluation Panel	NTU, Singapore
<i>Member</i>	<i>March 2021</i>
Tenure and Promotion Committee	SCSE, Singapore
<i>Member</i>	<i>1 Aug 2020</i>
Search Committee for Recruitment of Full Professor (Tenured) in AI	SCSE, Singapore
<i>Member</i>	<i>4 June 2020</i>
Search Committee for Recruitment of Full Professor (Practice)	SCSE, Singapore
<i>Member</i>	<i>15 Sep 2020</i>
NRF Fellow Evaluation Panel	NTU, Singapore
<i>Member</i>	<i>Sep 2020</i>
Continental-NTU Corporate Lab	NTU, Singapore
<i>Trust Lead</i>	<i>1 Dec 2019 - 31 Oct 2025</i>
HP-NTU Digital Manufacturing Corp Lab	NTU, Singapore
<i>Programme Director</i>	<i>1 Nov 2018 - 31 Oct 2022</i>
Energy Research Institute @NTU	NTU, Singapore
<i>Cluster Director in Cybersecurity</i>	<i>1 March 2019 - 31 Dec 2025</i>
Evaluation Committee for the Wallenberg for NTU PPF	NTU
<i>Committee Member</i>	<i>1 Dec 2018 – 30 Nov 2022</i>
Mentoring Committee	SCSE, NTU
<i>Mentoring Committee Member</i>	<i>1 Nov 2018 – 31 Oct 2022</i>
SAG Lead - Software Engineering	SCSE, NTU
<i>SAG Lead for Software Engineering</i>	<i>1 Sep 2018 – 31 Aug 2020</i>
NTU-Sweden WASP Workshop	NTU
<i>Organizer, NTU</i>	<i>4-5 Oct 2018</i>
Tier 2 Proposal Evaluation Committee	SCSE, NTU
<i>Tier 2 Proposal Evaluation Committee Member</i>	<i>1 Oct 2018</i>

Outreach Team of PhD programs for SCSE <i>Outreach Team Member</i>	SCSE, NTU <i>Jan 2017 - Now</i>
Research Group Lead for Cyber Security & Forensics <i>Research Group Lead</i>	SCSE, NTU <i>Nov 2017 - Now</i>
RSS Application Committee <i>RSS Application Committee Member</i>	SCSE, NTU <i>1 April 2016</i>
NTU CRP Panel for Informatics and Mathematics <i>Panel Member</i>	NTU <i>April 2013 - April 2015</i>
Singapore-Luxembourg Cybersecurity Workshop <i>Organizer, NTU</i>	NTU <i>2-3 March 2016</i>
Presenting NTU Research in SG-Israel Cyber-Security Workshop <i>Presenter, NTU</i>	NTU <i>29 Feb - 1 March 2016</i>
Presenting NTU Research to visit FireEye <i>Presenter, FireEye Headquarter</i>	NTU <i>16 Nov 2015</i>
Presenting NTU Research in SG-Israel Cyber-Security Workshop <i>Presenter, NTU</i>	NTU <i>23 Oct 2015</i>
Presenting NTU Research in NTU-Israel Cyber-Security Workshop <i>Presenter, in Tel Aviv University</i>	NTU <i>16 – 19 June 2015</i>
Presenting NTU Research in SG-UK Cyber-Security Workshop <i>Presenter, NTU</i>	NTU <i>24 Feb 2015–25 Feb 2015</i>
Presenting NTU Research to visit UC Berkeley <i>Presenter, in UC Berkeley</i>	NTU <i>4 July 2014–6 July 2014</i>
Presenting NTU Research in SG-UK Cyber-Security Workshop <i>Presenter, in Oxford University</i>	NTU <i>1 April 2014–7 April 2014</i>
(2) Service to Society/Industry.....	
National Cybersecurity R&D Technical Review Panel (TRP) <i>Core Member</i>	CSA, Singapore <i>1 Jan 2022 - 31 March 2026</i>
LLM Security Challenge <i>Technical Panel Chair</i>	AISG, Singapore <i>1 Oct 2024 - 31 Dec 2025</i>
GYSS 2022 Video Contest Evaluation Panel <i>Member</i>	GYSS, Singapore <i>Jan 2022</i>
ACM SIGSAC China Award Evaluation Panel <i>Member</i>	ACM, China <i>Sep 2021</i>
Cybersecurity Challenge Call 2022 <i>Technical Panel Member</i>	CSA, Singapore <i>1 Oct 2021 - 31 Dec 2022</i>
AI Security Standard Working Group <i>Chair</i>	AITC, Singapore <i>Jan 2021 - Dec 2022</i>
Ant Finance Cybersecurity Competition <i>Evaluation Panel</i>	China <i>April 2020 - Dec 2025</i>
National Satellite of Excellence in Mobile and Cloud Security <i>Evaluation Committee Member</i>	NRF, Singapore <i>1 Nov 2018 - 31 Oct 2021</i>

Cybersecurity Challenge Call 2019 <i>Technical Mentor</i>	CSA, Singapore <i>1 Aug 2019 - 31 Dec 2020</i>
Natural Sciences and Engineering Research Council of Canada <i>External Grant Proposal Review</i>	Canada <i>Dec 2019</i>
Ant Finance Cybersecurity Grant Call <i>Grant Review Panel</i>	China <i>Jan 2019</i>
Alibaba-NTU Corp Lab Grant Call <i>Grant Review Panel</i>	Singapore <i>Jan 2019</i>
TR68 Standard on Autonomous Vehicles Cybersecurity <i>Technical Committee Member</i>	Enterprise Singapore <i>1 Jan 2020 - 31 Dec 2021</i>
National Satellite of Excellence in Trustworthy Software Systems <i>Deputy Director</i>	NRF, Singapore <i>1 Nov 2018 - 31 Oct 2022</i>
Singapore Cybersecurity Consortium <i>Singapore Cybersecurity Consortium Management Committee</i>	NRF, Singapore <i>1 Sep 2016 to 31 Aug 2019</i>
One Day Course: Security for Automotous Vehicle <i>One Day Course for DSTA</i>	Singapore <i>April 21, 2016</i>
One Day Tutorial: Formal Methods and Its Applications to Cybersecurity <i>One Day Courses for Japanese Industry, hosted by AIST institute</i>	Japan <i>March 22, 2016</i>
One Day Course: Security Verification of the Exsection Stack <i>One Day Course for DSO</i>	Singapore <i>Oct 21, 2015</i>
One Day Course: Malware Detection and Binary Analysis <i>One Day Course for CSIT</i>	Singapore <i>April 2015</i>
Two Day Course: Introduction to Model Checking and its Applications <i>Two Day Courses for Japanese Industry, hosted by AIST institute</i>	Japan <i>Mar 20 to 23, 2015</i>
Half Day Tutorial on Security and Software Certification <i>Training Course for IDA</i>	Singapore <i>Oct 29th 2014</i>
One Day Tutorial: Introduction to PAT and its Applications <i>Training Course for Singapore Defence Researchers and Industries Engineers</i>	Singapore <i>Aug 21th 2013</i>
Half Day Tutorial: Build Your Own Model Checker in One Month <i>The 35th International Conference on Software Engineering</i>	San Francisco, USA <i>May 20th 2013</i>
One Week Course: Software Model Checking <i>Summer School Course in Tian Jin University</i>	Tian Jin University, China <i>July 2011</i>
Half Day Tutorial: Introduction to PAT <i>The 17th International Symposium on Formal Methods (FM 2011)</i>	Lero, Limerick, Ireland <i>June 20th, 2011</i>
Short Course on Automatic Verification of Real-time Systems <i>DSO National Laboratories</i>	Singapore <i>May 2011</i>
Short Course on Software Verification <i>ZheJiang University</i>	China <i>May 2011</i>

Commercialization

Technology Disclosure and Patent

1. TD-087-16: Methods For Vulnerability Detection In Programs (**Assigned at 75K SGD**).
2. TD-098-16: Semantics-Based Online Malware Detection: Towards Efficient Real-Time Protection Against Malware
3. TD-099-16: Malware Defense Using Attack Behavior Model
4. TD-100-16: Runtime Security Protection Using Hardware Specific Features
5. TD-109-16: Hardware Techniques For Runtime Anomaly Detection
6. TD-2018-199: Automated Mobile App Testing And Analysis
7. 2020-245-01-SG PRV: Marble: Model-Based Robustness Analysis of Stateful Deep Learning Systems (filed on 23 October 2020 and accorded Singapore provisional patent application number 10202010532Q)
8. TD-2021-001: Greybox fuzzing for Some/IP protocol in Automotive Ethernet (CONTINENTAL Ref: 2021E00365 SG)
9. TD-2021-010: Curiosity-based HMI Software Path Finder (CONTINENTAL Ref: 2021E00366 SG, **Patent filed with NTU ID 2021-010-01-DE**)
10. TD-2021-296: A Holistic DRL-Powered HMI Software Exploration Framework for Testing (**Patent filed with NTU ID 2021-296-01-DE**)
11. TD-2021-425: Methods and Apparatus for Automotive Software Trustworthiness Assessment
12. TD-2022-246: Evolutionary Test Case Generation Framework (CONTINENTAL Ref: 2022E03494 **Patent filed** under "METHOD AND SYSTEM FOR AUTOMATIC TEST PLAN GENERATION")
13. TD-2022-274: Control-Flow-Sensitive Fuzz Driver Generation (CONTINENTAL Ref: 2022E04146 SG, **Patent filed** under "A computer-implemented method for generating an automata-guided fuzz driver and fuzzing method using the fuzz driver")
14. TD-2022-383: Efficient Greybox fuzzing for Automotive Grade Linux (AGL) Applications (CONTINENTAL Ref: 2022E05936 SG).
15. TD-2022-449: Efficient Root Cause Analysis Based on Crash Clustering (CONTINENTAL Ref: 2022E06879 SG)
16. TD-2023-037: A Hardware/Software Collaborative Communication Channel for Trusted Execution Environment (CONTINENTAL Ref: 2023E00530 SG)
17. TD-2023-039: Automated Testing for Exposing Resource Exhaustion DoS Vulnerabilities (CONTINENTAL Ref: 2023E00603-SG **US Patent granted WO2025045976A1, U.S., 2025.**)
18. TD-2023-049: Interpretability-Based EEG Channel Selection Tool for Driver Drowsiness Detection
19. TD-2023-080: Synthesizing Efficiently with Derivation Pattern Mining

20. TD-2023-098: Scenario Semantics-Driven Methods for Autonomous Driving System Testing (Assigned at 150K SGD)
21. TD-2023-262: A More Efficient Fuzzing Strategy For Automated Root Cause Analysis And Explanation (CONTINENTAL Ref: 2023E03708 SG)
22. TD-2024-022: An Enhanced AFL-Based Greybox Fuzzing Approach For Rehosting Embedded Applications (CONTINENTAL Ref: 2024E00350 SG, Patent filed)
23. TD-2024-089: PatchSleuth: Sleuthing Patch Commits for Disclosed Vulnerabilities in Open-Source Software.
24. TD-2024-224: BootHotPatch: Transparent Hot Patch Automatic Deployment Framework during Linux-Based Vehicle System Boot. (CONTINENTAL ref: 2024E03625 SG, Patent filed)
25. TD-2024-241: Fuzzing4AllLib: Generating Fuzz Testing Drivers Using the Coding Capabilities of Large Language Models. (CONTINENTAL ref: 2024E04082 SG, Patent filed)
26. TD-2024-243: AutoPatcher: Automatic Root-Cause-Analysis Guided Program Repair via Large Language Models. (CONTINENTAL ref: 2024E04091 SG, Patent filed)
27. TD-2024-245: LLM4RCA: LLM-Based Multi-Agent Root Cause Analysis. (CONTINENTAL ref: 2024E04182 SG, Patent filed)
28. TD-2025-125: ARASH: Analysis and Repair of Crash. (CONTINENTAL ref: 2025E01753 SG, Patent filed)
29. TD-2025-128: Automatic Java Deserialization Vulnerability Mining and Exploitation via LLM Agents. (CONTINENTAL ref: 2025E01925 SG, Patent filed)
30. TD-2025-345: LogicFuzz: Coordinated LLM Agents for Next-Gen Fuzz Driver Generation (CONTINENTAL ref: 2025E04486 SG, Patent filed)
31. TD-2025-391: Synth: Synthesizing Static Analysis Rules for Vulnerability Detection via Semantic Mining". (CONTINENTAL ref: 2025E01925 SG, Patent filed)
32. TD-2025-436: PGFilter: A parameter-guided, token-saving source code filter for LLM-based root cause analysis (CONTINENTAL ref: 2025E05430 SG, Patent filed)
33. TD-2025-497: A Knowledge-Guided Multi-Agent Framework for Intelligent Software Development.
34. TD-2025-502: A Brain-Inspired Embodied Robotic Agent Based On Memory System (Assigned at 120K SGD at March 2026).
35. TD-2025-505: Secure Home Care Agent (Assigned at 120K SGD at Jan 2026).
36. TP305: An Optimization-based Jailbreak Attack System for Large Language Models (Assigned at 200K RMB at Feb 2026).
37. TD-2025-526: A Cross-Modal Memory-Driven Medical Semantic Reasoning Agents.
38. TD-2026-001: Brainary - A Cognitive AI Agent Computing Architecture.

39. TD-2026-002: Neuro-Symbolic Detection for C/C++ Path-Sensitive Vulnerabilities (Aumovio's Reference: 2026E00104 SG, **Patent filed**)
40. TD-2026-026: Software Gene Platform Centre

Entrepreneurship

- 1) I co-founded Semantic Engineering Pte Ltd with my Ph.D. supervisor in 2007 in IT industry. The main product is a model checking software named PATPro, which is to provide the formal verification of software systems for correctness and security. PATPro is well recognized in the industry, especially in Japan. Many advanced industry research labs have purchased PAT Pro, e.g., CATS, AutoNetworks Technologies, National Institute of Advanced Industrial Science and Technology, NTT, CAV-Tech, Mitsubishi Space Software Co., Japan Aerospace eXploration Agency, BD Biosciences, etc. With the help of the Japanese distributor, the total sales in last 3 years are 1.1 million SGD.
- 2) We have developed an app testing and vulnerability analysis platform. Due to the direct impact of this line of research, we received the GAP funding of 250k SGD from NRF to commercialize the IP (TD-2018-199: Automated Mobile App Testing and Analysis) as a cloud based security assessment service in Dec 2017. We have finished the first release of the product development and look for potential customer currently.
- 3) I co-founded Scantist Pte. Ltd. In 2017 in cybersecurity industry. The main solution is to provide automatic vulnerability detection and management services for source code and binary. We have filed one technical disclosure (TD-087-16: Methods for Vulnerability Detection in Programs) with NTUitive, and has been successfully assigned to Scantist with 75k SGD assignment fee. So far we have received 10 million SGD VC investment and setup a 30 strong research team in the company and current valuation is 40 million USD. The customers of our product include various universities including NTU, IT companies like Huawei, banking industry, automotive industry and insurance companies. The sales is 3+ million SGD revenue in last 12 months.

PUBLICATIONS

Note that: * indicates students under my supervision, ** for researchers in my group by the time of submission, and ^ indicates that I am the corresponding author.

[##] indicates the top tier **A+** conference/journal.

[#] indicates the top tier **A** conference/journal.

Bold name **Yang Liu** implies that I am the main author: i) first author, ii) corresponding author, or iii) main supervisor of the student/researcher who is first author.

Journal Papers

1. [##1A] Yuanzhou Yuan, Wenbing Tang**, Mingfeng Cheng, Renzhi Wang, Chengwei Liu**, Yang Liu, Huading Zou. "Causality-aware Safety Testing for Autonomous Driving Systems," IEEE Transactions on Software Engineering (TSE), 2026. (IF 7.4)
2. [##1A] Jian Zhang**, Cong Wang**, Anran Li**, Weisong Sun**, Cen Zhang**, **Yang Liu**. *Evaluating Large Language Models for Line-Level Vulnerability Localization* IEEE Transactions on Software Engineering (TSE)

3. [##1B] Cuiying Gao**, Hanwen Zhang, Heng Li, Wei Zhou, Haolan Zhang, Yueming Wu**, Wei Yuan, **Yang Liu**. "Seeking Certainty from Uncertainty: A Training Framework for Android Malware Detection Resilient to Label Noise," ACM Transactions on Software Engineering and Methodology (TOSEM), 2025. (IF 6.2)
4. [##1A] Qi Zhou, Dongxia Wang, Tianlin Li**, Zhihong Xu, Yang Liu, Kui Ren, Wenhai Wang, Qing Guo, "FOOLSDEDIT: Deceptively Steering Your Edits Towards Targeted Attribute-aware Distributio," IEEE Transactions on Dependable and Secure Computing (TDSC), 2025. (IF 7.3)
5. [##1A] Wenting Zhao, Wuxia Jin, Yiran Zhang**, Ming Fan, Haijun Wang, Li Li, Yang Liu, Ting Liu, "Software Architecture Recovery Augmented with Semantic," IEEE Transactions on Software Engineering (TSE), 2025. (IF 7.4)
6. [##1A] Cong Wu**, Hangcheng Cao, Jing Chen, Guowen Xu, Ziming Zhao, **Yang Liu**, Hongbo Jiang, "RUGSCREENER: Leveraging Temporal Graph Neural Network for Rugpull Detection in DeFi," IEEE Transactions on Information Forensics & Security (TIFS), 2025. (IF 6.8)
7. [##1A] Jianguo Zhao, Yuqiang Sun*, Cheng Huang, Chengwei Liu**, YaoHui Guan, Yutong Zeng, Yang Liu, "Towards Secure Code Generation with LLMs: A Study on Common Weakness Enumeration," IEEE Transactions on Software Engineering (TSE), 2025. (IF 7.4)
8. [##1A] Shangqing Liu**, Daya Guo, Jian Zhang**, Wei Ma**, Yanzhou Li*, **Yang Liu**, "An Empirical Study of Exploring the Capabilities of Large Language Models in Code Learning," IEEE Transactions on Software Engineering (TSE), 2025. (IF 7.4)
9. [##1A] Kun Sun, Wei Ma**, Cong Wu**, Jing Chen, Lingxiao Jiang, Shangqing Liu**, Yang Liu, "Detecting DeFi Fraud with a Graph-Transformer Language Model," IEEE Transactions on Information Forensics & Security (TIFS), 2025. (IF 6.8)
10. [##1A] Qiao Li, Cong Wu**, Jing Chen, Zijun Zhang, Kun He, Ruiying Du, Xinxin Wang, Qingchuan Zhao, Yang Liu, "Privacy-preserving Universal Adversarial Defense for Black-box Models," IEEE Transactions on Information Forensics & Security (TIFS), 2025. (IF 6.8)
11. [##1B] Shihan Dou, Huiyuan Zheng, Junjie Shan, Yueming Wu**, Deqing Zou, Xuanjing Huang, Yang Liu, "A Scalable Vulnerability Detection System with Multi-View Graph Representations," ACM Transactions on Software Engineering and Methodology (TOSEM), 2025. (IF 6.2)
12. [##1A] Zhiyuan Wei, Jing Sun, Yuqiang Sun*, Ye Liu, Daoyuan Wu, ZiJian Zhang**, Xi-anhao Zhang, Meng Li, Yang Liu, Chunmiao Li, Mingchao Wan, Jin Dong, Liehuang Zhu, "Advanced Smart Contract Vulnerability Detection via LLM-Powered Multi-Agent Systems," IEEE Transactions on Software Engineering (TSE), 2025. (IF 7.4)
13. [##1A] Teng Li, Zejian Lin, Yebo Feng**, Chong Wang**, Zhuo Ma, Bin Xiao, Jianfeng Ma, Yang Liu, CryptlF: Towards Cloud-based IoT Anomaly Detection over Encrypted Feature Streams," IEEE Transactions on Knowledge and Data Engineering (TKDE), 2025. (IF 10.4)
14. [##1A] Lyuye Zhang**, Kaixuan Li**, Kairan Sun*, Daoyuan Wu**, Ye Liu**, Haoye Tian**, **Yang Liu**. *ACFix: Guiding LLMs with Mined Common RBAC Practices for Context-Aware Repair of Access Control Vulnerabilities in Smart Contracts*, IEEE Transactions on Software Engineering (TSE), vol. 51, no. 9, pp. 2512–2532, 2025. (IF 7.4)

15. [##1A] Cong Wu**, Jing Chen, Jiahong Li, Jiahua Xu, Ju Jia, Yutao Hu, Yebo Feng**, **Yang Liu**, Yang Xiang. *Profit or Deceit? Mitigating Pump and Dump in DeFi via Graph and Contrastive Learning*, IEEE Transactions on Information Forensics & Security (TIFS), vol. 20, pp. 8994–9008, 2025. (IF 6.8)
16. [##1A] Xiaojun Jia**, Sensen Gao, Qing Guo, Ke Ma, Yihao Huang**, Simeng Qin, **Yang Liu**, Ivor Tsang, Xiaochun Cao. *Semantic-Aligned Adversarial Evolution Triangle for High-Transferability Vision-Language Attack*, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), vol. 47, no. 10, pp. 8489–8505, 2025. (IF 23.6)
17. [##1A] Yueming Wu**, Wenqi Suo, Siyue Feng, Deqing Zou, Wei Yang, **Yang Liu**, Hai Jin. *MalScan: Android Malware Detection Based on Social-Network Centrality Analysis*, IEEE Transactions on Dependable and Secure Computing (TDSC), vol. 22, no. 5, pp. 5464–5479, 2025. (IF 7.3)
18. [##1B] Yuchen Chen, Weisong Sun**, Chunrong Fang, Zhenpeng Chen**, Yifei Ge, Tingxu Han, Quanjun Zhang, Yang Liu, Zhenyu Chen, Baowen Xu, Yang Liu, “Security of Language Models for Code: A Systematic Literature Review,” ACM Transactions on Software Engineering and Methodology (TOSEM), 2025. (IF 6.2)
19. [##1A] Jingxuan Chen, Dianrun Huang, Yijie Wang, Zipping Yu, Zhongliang Zhao, Xianbin Cao, Yang Liu, Tony Q.S. Quek. *Enhancing Routing Performance through Trajectory Planning with DRL in UAV-aided VANETs*, IEEE Transactions on Machine Learning in Communications and Networking (TMLCN), vol. 3, pp. 517–533, 2025.
20. [##1B] Jingquan Ge**, Yaowen Zheng**, Yuekang Li**, Wei Ma**, Sheikh Mahbub Habib, Praveen Kakkolangara, Gabriel Wayne Byman, **Yang Liu**, “OptRCA: A More Efficient and Accurate Approach for Automated Root Cause Analysis and Explanation,” ACM Transactions on Software Engineering and Methodology (TOSEM), 2025. (IF 6.2)
21. [##1B] Xiaoyu Zhang*, Cen Zhang**, Tianlin Li**, Yihao Huang**, Xiaojun Jia**, Ming Hu, Jie Zhang, **Yang Liu**, Shiqing Ma, Chao Shen, “JailGuard: A Universal Detection Framework for Prompt-based Attacks on LLM Systems,” ACM Transactions on Software Engineering and Methodology (TOSEM), 2025. (IF 6.2)
22. [##1A] Chenghao Li, Yifei Wu, Wenbo Shen, Zichen Zhao, Rui Chang, Chengwei Liu**, Yang Liu. *Demystifying Rust Unstable Features at Ecosystem Scale: Evolution, Propagation, and Mitigation*, IEEE Transactions on Software Engineering (TSE), vol. 51, no. 4, pp. 1284–1302, 2025. (IF 7.4)
23. [##1A] Xiyue Zhang*, Xiaoyong Xue, Xiaoning Du, Xiaofei Xie, **Yang Liu**, Meng Sun. *Runtime Backdoor Detection for Federated Learning via Representational Dissimilarity Analysis*, IEEE Transactions on Dependable and Secure Computing (TDSC), vol. 22, no. 5, pp. 4607–4624, 2025. (IF 7.3)
24. [##1A] Liming Zhai**, Lina Wang, Yanzhen Ren, **Yang Liu**. *Generalized Local Optimality for Video Steganalysis in Motion Vector Domain*, IEEE Transactions on Dependable and Secure Computing (TDSC), vol. 22, no. 4, pp. 4231–4247, 2025. (IF 7.3)

25. [##1A+] Yebo Feng**, Jun Li, Jelena Mirkovic, Cong Wu**, Chong Wang**, Hao Ren, Jiahua Xu, **Yang Liu**, "Unmasking the Internet: A Survey of Fine-Grained Network Traffic Analysis," IEEE Communications Surveys and Tutorials (COMST), 2025. (IF 34.4)
26. [##1A] Shide Zhou, Tianlin Li**, Yihao Huang**, Ling Shi**, Kailong Wang, Yang Liu, Haoyu Wang, "NeuSemSlice: Towards Effective DNN Model Maintenance via Neuron-level Semantic Slicing," ACM Transactions on Software Engineering and Methodology (TOSEM), 2025. (IF 6.2)
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 273. Lingyun Situ*, Linzhang Wang, **Yang Liu**, Bing Mao, Xuandong Li. *Vanguard: Detecting Missing Checks for Prognosing Potential Vulnerabilities*, the Tenth Asia-Pacific Symposium on Internetwork (Internetwork 2018), Article No. 5, Beijing, China, Sept. 2018. **Best Paper Award.**
 274. [##A+] Lingling Fan**, Ting Su**, Sen Chen**, Guozhu Meng**, **Yang Liu**, Lihua Xu and Geguang Pu. *Efficiently Manifesting Asynchronous Programming Errors in Android Apps*, The 33rd IEEE/ACM International Conference on Automated Software Engineering (ASE 2018), pp. 486–497, Montpellier, France 3 to 7 September, 2018. (Acceptance rate: $24\% = 83 / 346$)
 275. [##A+] Kaifeng Huang, Bihuan Chen**, Xin Peng, Daihong Zhou, Ying Wang, Yang Liu, and Wenyun Zhao. *CLDIFF: Generating Concise Linked Code Differences*, The 33rd IEEE/ACM International Conference on Automated Software Engineering (ASE 2018), pp. 679–690, Montpellier, France 3 to 7 September, 2018. (Acceptance rate: $24\% = 83 / 346$) **ACM Distinguished Paper Award.**
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279. **[#A]** Hongxu Chen*, Alwen Tiu, Zhiwu Xu and **Yang Liu**, *A Permission-Dependent Type System for Secure Information Flow Analysis*, The 31st IEEE Computer Security Foundations Symposium (CSF 2018), pp. 218–232, Oxford, UK, July 9-12, 2018. (Acceptance rate: 34% = 25 /72)
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282. **[##A+]** Lingling Fan**, Ting Su**, Sen Chen**, Guozhu Meng**, **Yang Liu**, Lihua Xu, Geguang Pu and Zhendong Su. *Large-Scale Analysis of Framework-Specific Exceptions in Android Apps*. The 40th International Conference on Software Engineering (ICSE 2018), pp. 408–419, Gothenburg, Sweden, May 27-3 Jun., 2018. (Acceptance rate: 20.9% = 105/502) **ACM Distinguished Paper Award**.
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Note: Zhenchang Xing and Yang Liu have equal contribution in this work.
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286. **[##A+]** Shang-Wei Lin, Jun Sun, Hao Xiao*, Yang Liu, David Sanan and Henri Hansen. *FiB: Squeezing Loop Invariants by Interpolation between Forward/Backward Predicate Transformers*.

The 32nd IEEE/ACM International Conference on Automated Software Engineering (ASE 2017), pp. 793–803, Urbana-Champaign, IL, USA. Oct. 30–Nov. 3, 2017. (Acceptance rate: 20.7% = 65/314) **ACM Distinguished Paper Award**.

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- Note: Shang-Wei Lin and Yang Liu have equal contribution in this work.
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- Note: Shang-Wei Lin and Yang Liu have equal contribution in this work.
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300. Xiaojun Wang, Hesuan Hu**, Yuan Zhou*, and Yang Liu, *A robust control approach to automated manufacturing systems allowing failures and reworks with Petri nets*, in Proceedings of IEEE International Conference on Automation Science and Engineering (CASE 2017), pp. 370-375, Xi'an, China, August 2017.
301. Nan Du, Hesuan Hu**, Yuan Zhou*, and Yang Liu, *Robust control of automated manufacturing systems with complex structures using Petri nets*, in Proceedings of IEEE International Conference on Automation Science and Engineering (CASE 2017), pp. 364-369, Xi'an, China, August 2017.
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314. **[##A+]** Guozhu Meng*, Yinxing Xue**, Mahinthan Chandramohan*, Annamalai Narayanan**, **Yang Liu**, Jie Zhang, and Tieming Chen. *Mystique: Evolving Android Malware for Auditing Anti-Malware Tools*. 2016 ACM Asia Conference on Computer and Communications Security (ASIACCS 2016), pp. 365–376, Xi'an, China, May 30–Jun. 3, 2016. (Acceptance rate: 73/350=20.9%).
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316. **[##A+]** Bihuan Chen**, **Yang Liu**, and Wei Le. *Generating Performance Distributions via Probabilistic Symbolic Execution*. The 38th International Conference on Software Engineering (ICSE 2016), pp. 49–60, Austin, TX, USA, May 14–22, 2016. (Acceptance rate: 19%).
317. **[##A+]** Tian Huat Tan, Manman Chen, Jun Sun, Yang Liu, Étienne André, Yinxing Xue, and Jin Song Dong. *Optimizing Selection of Competing Services with Probabilistic Hierarchical Refinement*. The 38th International Conference on Software Engineering (ICSE 2016), pp. 85–95, Austin, TX, USA, May 14–22, 2016. (Acceptance rate: 19%).

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Note: Jie Zhang and Yang Liu have equal contribution in this work.
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322. Nan Du, Hesuan Hu**, Yang Liu. *Robust control of automated manufacturing systems with assembly operations using petri nets*. The IEEE International Conference on Robotics and Automation (ICRA 2016), pp. 3632–3638, Stockholm, Sweden, May 16-21, 2016.
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324. Hesuan Hu**, Yan Yang, **Yang Liu**, and Nan Du. *Critical Stages and Their Application in Large Scale Automated Manufacturing Systems Via Petri Nets*. The 15th European Control Conference (ECC 2016), pp. 2337–2344, Aalborg, Denmark, June 29-July 1, 2016.
325. **[##A+]** Shang-Wei Lin, Truong Khanh Nguyen, Jun Sun, Yang Liu, and Jin Song Dong. *Interpolation Guided Compositional Verification*. The 30th IEEE/ACM International Conference on Automated Software Engineering (ASE 2015), pp. 65–74, Lincoln, Nebraska, USA, Nov. 9-13, 2015. (Acceptance rate: 20.8%=60/289)
326. **[##A+]** Jun Sun, Hao Xiao*, Yang Liu, Shang-Wei Lin, and Shengchao Qin. *TLV: Abstraction through Testing, Learning and Validation*. The 23rd ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE 2015), pp. 698–709, Bergamo, Italy, Aug. 30-Sept. 4, 2015. (Acceptance rate: 25.4%=74/291)
327. **[#A]** Songzheng Song**, Jiexin Zhang*, **Yang Liu**, Mikhail Auguston, Jun Sun, Jin Song Dong, and Tieming Chen. *Formalizing and Verifying Stochastic System Architectures Using Monterey Phoenix (SoSyM Abstract)*. ACM/IEEE 18th International Conference on Model Driven Engineering Languages and Systems (MODELS 2015), p. 449, Ottawa, ON, Canada, Sept. 27-Oct. 2, 2015. **Best Paper Award**.

328. [##A+] Dongxia Wang*, Tim Muller**, Jie Zhang, and **Yang Liu**. *Quantifying Robustness of Trust Systems Against Collusive Unfair Rating Attacks Using Information Theory*. The 25th International Joint Conference on Artificial Intelligence (IJCAI 2015), pp. 111–117, Buenos Aires, Argentina, Jul. 25–31, 2015. (Acceptance rate: 28.8%=575 /1996)
Note: Jie Zhang and Yang Liu have equal contribution in this work.
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335. [##A+] Xiaofei Xie**, **Yang Liu**, Wei Le, Xiaohong Li, and Hongxu Chen. *S-Looper: Automatic Summarization for Multipath String Loops*. International Symposium on Software Testing and Analysis (ISSTA 2015), pp. 188–198, Baltimore, MD, USA, Jul. 12–17, 2015. (Acceptance rate 27.7%=33/117)
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339. [##A+] Li Li, Jun Sun, Yang Liu, and Jin Song Dong. *Verifying Parameterized Timed Security Protocols*. The 20th International Symposium on Formal Methods (FM 2015), pp. 342–359, Oslo, Norway, Jun. 22–26, 2015. (Acceptance rate: 26%)
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Note: Jie Zhang and Yang Liu have equal contribution in this work.
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347. Yan Yang, Hesuan Hu**, Yang Liu. *A Petri net-based distributed control of automated manufacturing systems with assembly operations*. IEEE International Conference on Automation Science and Engineering (CASE 2015), pp. 1090–1097, Gothenburg, Sweden, Aug. 24–28, 2015.
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Book Chapter

1. Jun Sun, Yang Liu and Jin Song Dong, *Timed Automata*. Handbook of Finite State Based Models and Applications, Jiachun Wang (Editor), Boca Raton, FL, USA: CRC Press, 2012.

Keynote Address

1. *A Road Towards an Interaction between Cyber Security and AIGC*, 32nd Asia-Pacific Software Engineering Conference (APSEC 2025), Dec, 2025.
2. *From Software Composition Analysis to Open Source Management*, The 5th Conference on Safety Critical Software Testing (CSCST), Dec, 2021.
3. *The Journey of Fuzzing Research*, The 4th International Conference on Mutation Testing (MT 2021), Oct, 2021.
4. *A Road Towards an Interaction between Cyber Security and Artificial Intelligence*, The 16th International Conference on Information Security and Cryptology (INSCRYPT 2020), December, 2020.
5. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, The 4th Working Formal Methods Symposium (FROM 2020), September, 2020.
6. *Security Verification: Attempts and Challenges*, The 9th International Workshop on Security Proofs for Embedded Systems (PROOFS), September, 2020.
7. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, YEF2020, June, 2020, Beijing, China.
8. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, International Symposium on Theoretical Aspects of Software Engineering (TASE 2020), July, 2020, Hangzhou, China.
9. *Taming Vulnerabilities to Provide Better Software Supply Chain Security*, the International Conference on Dependable Systems and Their Applications (DSA 2019) Harbin, China, Jan 03-06, 2020.
10. *Detect and Manage the Vulnerabilities in the Open Source Software*, the 18th International Conference on Software and Systems Reuse (ICSR 2019) Cincinnati, OH, United States, June 26-28, 2019.

11. *Secure Deep Learning Engineering: A Road Towards Quality Assurance of Intelligent Systems*, The 21st International Conference on Formal Engineering Methods (ICFEM 2019), November 5th-9th, 2019, Shenzhen, China.
12. *Cyber-security: the Journey from Formal Methods, Program Analysis, Data Analytics to Artificial Intelligence*, the Tenth Asia-Pacific Symposium on Internetware (Internetware 2018), Beijing, China, September, 2018.
13. *The Unknown Journey of Academics*, The 16th National Software Application Conference (NASAC 2018), Shenzheng, China, Nov, 2018.
14. *Cyber-security, The Journey from Formal Methods, Program Analysis to Data Analytics*, The 22nd International Conference on Engineering of Complex Computer Systems (ICECCS 2017), Fukuoka, Japan, November 6-8, 2017.
15. *Our Journey on Search-based SE Research: from FM to security*, The 5th Asian-Pacific Workshop of Advanced Software Engineering (AWASE 2016), 19-20 March 2016, Nara, Japan.
16. *Event and Strategy Analytics*. 2015 International Symposium on Theoretical Aspects of Software Engineering (TASE), Nanjing, China, September 12-14, 2015.
17. *Event Analytics*. The 11th International Colloquium on Theoretical Aspects of Computing (ICTAC 2014), Bucharest, Romania, Sept. 17-19, 2014.

Editorship

1. Associate Editor, *IEEE Transactions on Dependable and Secure Computing (TDSC)*. (IF 7.3)
2. Associate Editor, *Machine Intelligence Research (MIR)*. (IF 8.7)
3. Associate Editor, *Cybersecurity*. (IF 3.7)
4. Associate Editor, *IEEE Transactions on Information Forensics and Security (TIFS)*. (IF 6.8)
5. Guest Editor of Special Issue On Reliability and Robustness in AI-Based Cybersecurity Solution, *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2022. (IF 7.3)
6. Associate Editor, *ACM Transactions on Software Engineering and Methodology (TOSEM)*, since 2022. (IF 6.2)
7. Associate Editor, *IEEE Access Journal*, since 2014, Impact factor 3.557.
8. Associate Editor, *Journal of Computer Science*, since 2014.
9. Editorial Advisory Board Member, *Book: Overcoming Challenges in Software Engineering Education: Delivering Non-Technical Knowledge and Skills*, 2013.

Publication Statistics Summary

My publications include 8 homerun papers ($FWCI \geq 20$) and 1 breakthrough paper (with more than 6 times the citation growth between years 4-20 post-publication compared to years 0-3).

My publications appears in top tier conferences (e.g., 53 ICSE, 25 FSE, 44 ASE, 21 ISSTA, 4 CAV, 13 FM, 3 TACAS, 8 S&P(Oakland), 8 CCS, 12 Usenix Security, 5 NDSS, 15 AAI, 6 IJCAI, 4 AAMAS, 12 ICLR and 7 NeurIPS), and top tier journals (e.g., 32 TSE, 26 TOSEM, 24 TIFS, 13 TDSC, 3 TPAMI, 5 ACM Computing Survey, 3 TCST, 1 TOPLAS, 2 Automatica, 2 TIP and 3 TMC).

All Years	Papers	Main Author	Tier 1			Tier 1 Main Author		
			A+	A	B	A+	A	B
Total	641	405	501			345		
Journal	193	121	20	119	27	11	82	17
			166			110		
Conference	448	284	302	33		210	25	
			335			235		
Book	6							
Book Chapter	1							

Last 3 Years	Papers	Main Author	Tier 1			Tier 1 Main Author		
			A+	A	B	A+	A	B
Total	234	147	233			147		
Journal	82	46	11	52	18	6	29	11
			81			46		
Conference	152	101	134	18		89	12	
			152			101		

Last Year	Papers	Main Author	Tier 1			Tier 1 Main Author		
			A+	A	B	A+	A	B
Total	120	72	120			72		
Journal	36	19	3	27	6	1	14	4
			36			19		
Conference	84	53	78	6		50	3	
			84			53		